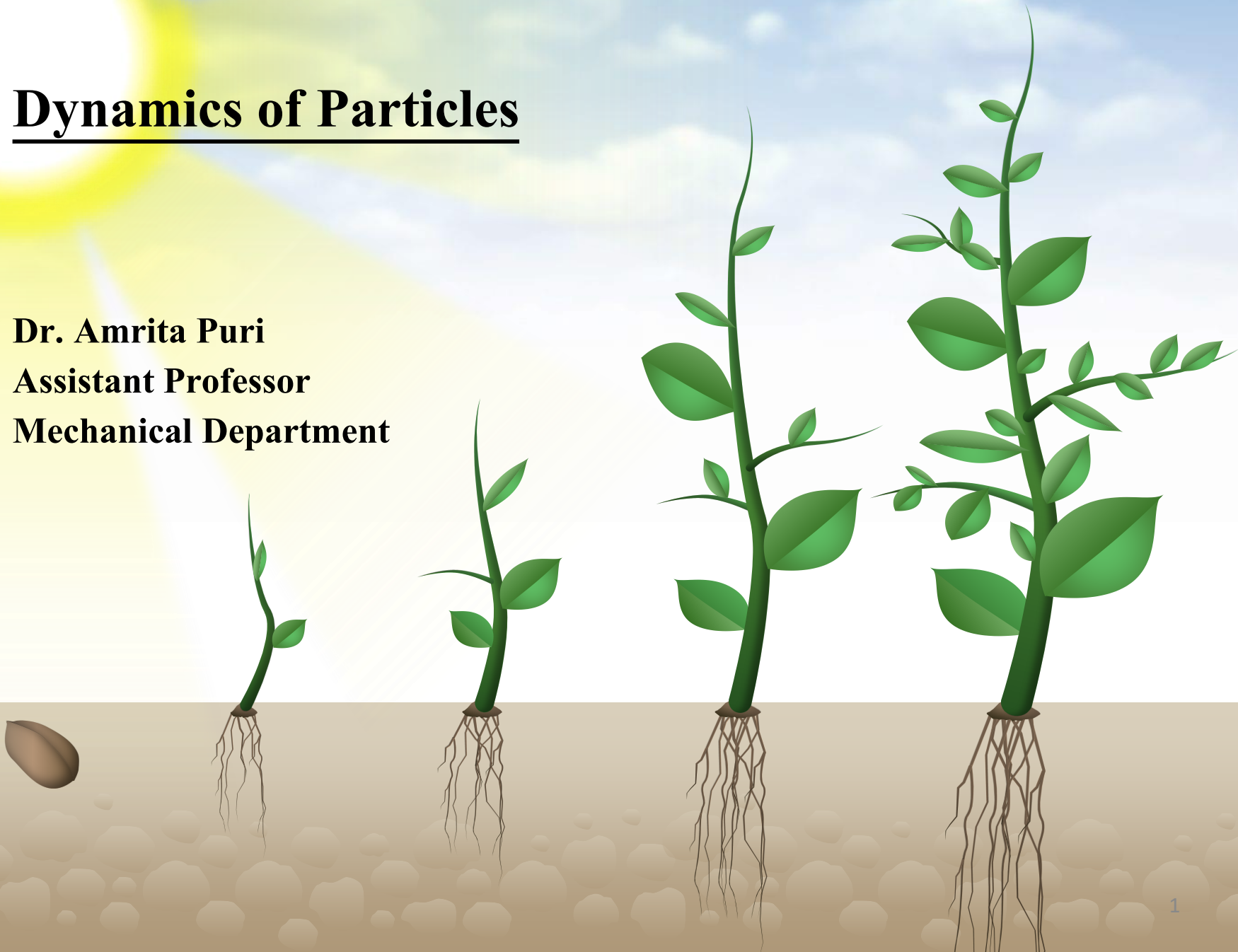


Dynamics of Particles

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Dynamics of Particles

Dynamics of Particles [6 Lectures]:

Chapter 2 [Meriam and Kraige]

Kinematics of particles, Velocity and acceleration in terms of path variables, simple relative motion, motion of particle relative to a pair of translating axes

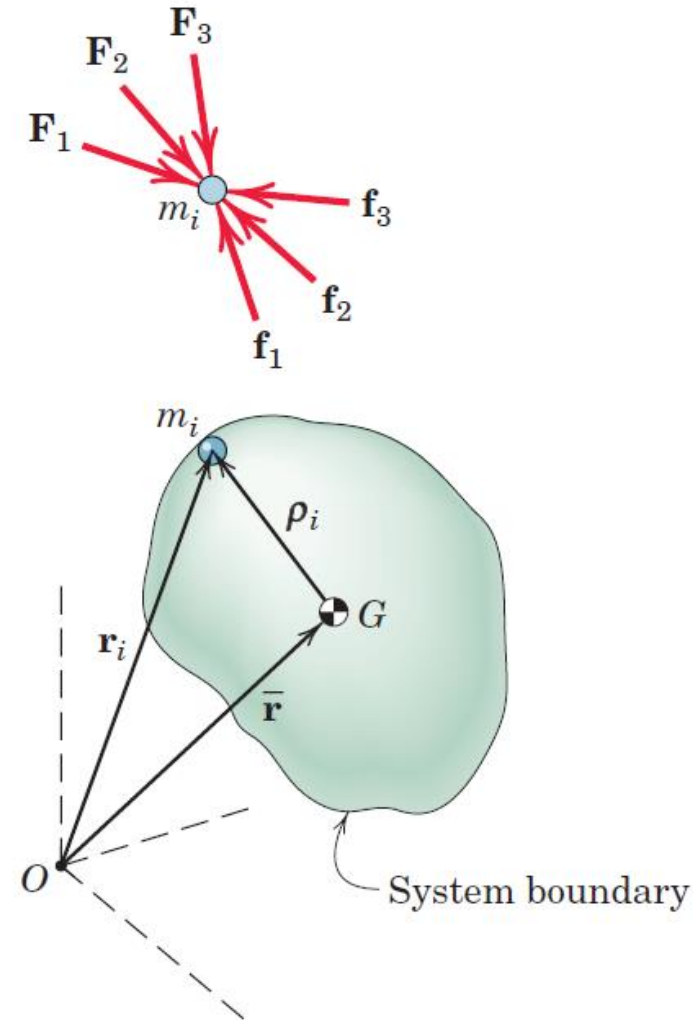
Chapter 3 (single particle) and Chapter 4 (system of particles) [Meriam and Kraige]

Newton's laws of rectangular coordinates/rectilinear translation, cylindrical coordinate/Central force motion, Conservation of Mechanical Energy, Work-energy equations, Center of mass based Kinetic energy, Impulse and Momentum relation of particles, Moment of momentum equations - single particle/system of particles

Generalized Newton's Second Law

General Mass System: A system of n mass particles enveloped by a closed surface in space

- Exterior surface of a given rigid body
- Bounding surface of an arbitrary portion of the body
- Exterior surface of a rocket containing both rigid and flowing particles
- A particular volume of fluid particles



Generalized Newton's Second Law

External forces (outside the envelope)
acting on particle m_i

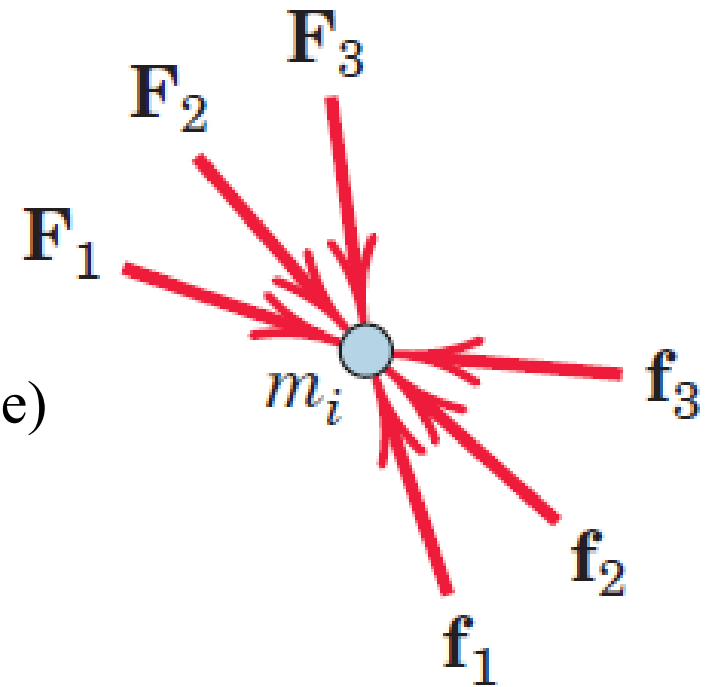
$F_1, F_2, F_3, F_4 \dots\dots\dots$

Internal reaction forces (inside the envelope)
acting on particle m_i

$f_1, f_2, f_3, f_4 \dots\dots\dots$

Newton's second law applied on m_i

$$F_1 + F_2 + F_3 + \dots + f_1 + f_2 + f_3 = m_i \ddot{r}_i$$



Generalized Newton's Second Law

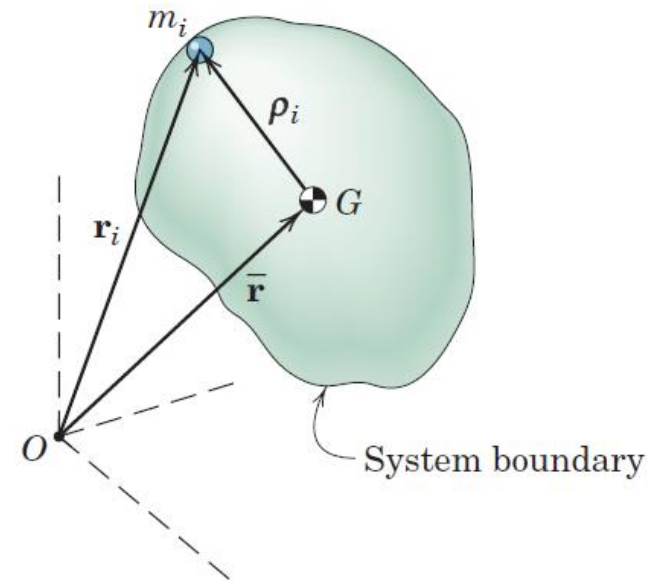
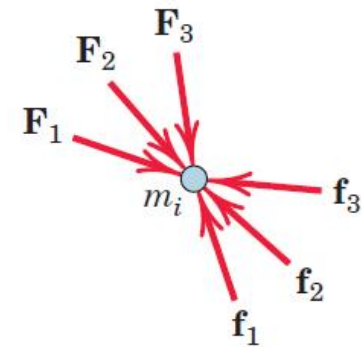
Position Vector of Mass Centre G

$$m\bar{r} = \sum_{i=1}^n m_i r_i \quad m = \sum_{i=1}^n m_i$$

Sum for all n mass particles

$$\sum F + \sum f = \sum_{i=1}^n m_i \ddot{r}_i$$

$$\sum F = \sum_{i=1}^n m_i \ddot{r}_i$$



Generalized Newton's Second Law

Position Vector of Mass Centre G

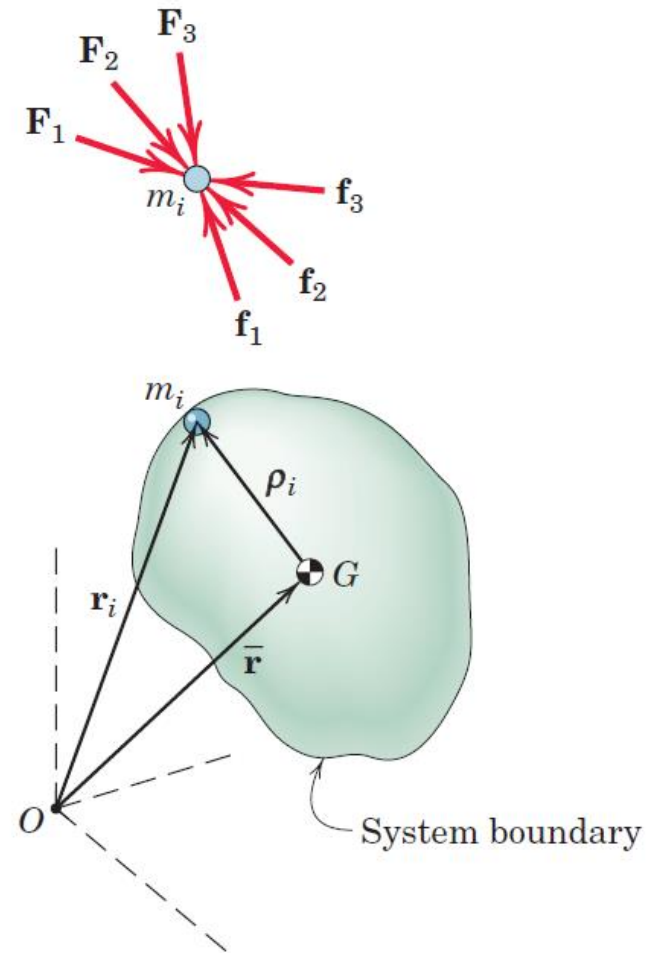
$$m\bar{\mathbf{r}} = \sum_{i=1}^n m_i \mathbf{r}_i$$

$$m\ddot{\mathbf{r}} = \sum_{i=1}^n m_i \ddot{\mathbf{r}}_i$$

Sum for all n mass particles

$$\sum F = \sum_{i=1}^n m_i \ddot{\mathbf{r}}_i = m\ddot{\mathbf{r}} = m\bar{\mathbf{a}}$$

$\bar{\mathbf{a}}$ is acceleration of mass center of the body



Generalized Newton's Second Law

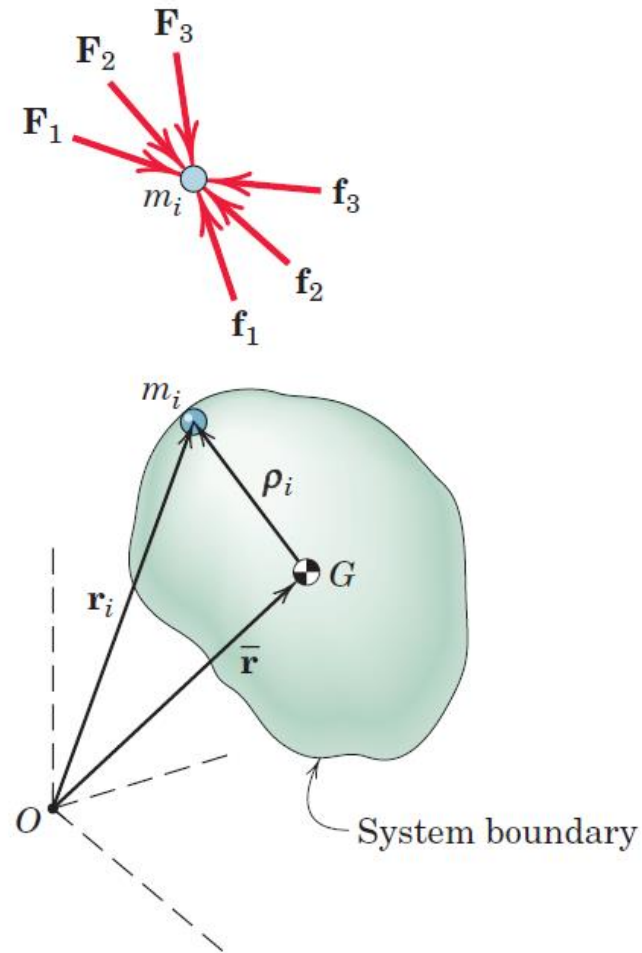
Sum for all n mass particles

$$\sum F = \sum_{i=1}^n m_i \ddot{\mathbf{r}}_i = m \ddot{\mathbf{r}} = m \bar{\mathbf{a}}$$

Instantaneous relationship

$\bar{\mathbf{a}}$ is acceleration of mass center of the body

Resultant of the external forces on *any* system of masses equals the total mass of the system times the acceleration of the center of mass



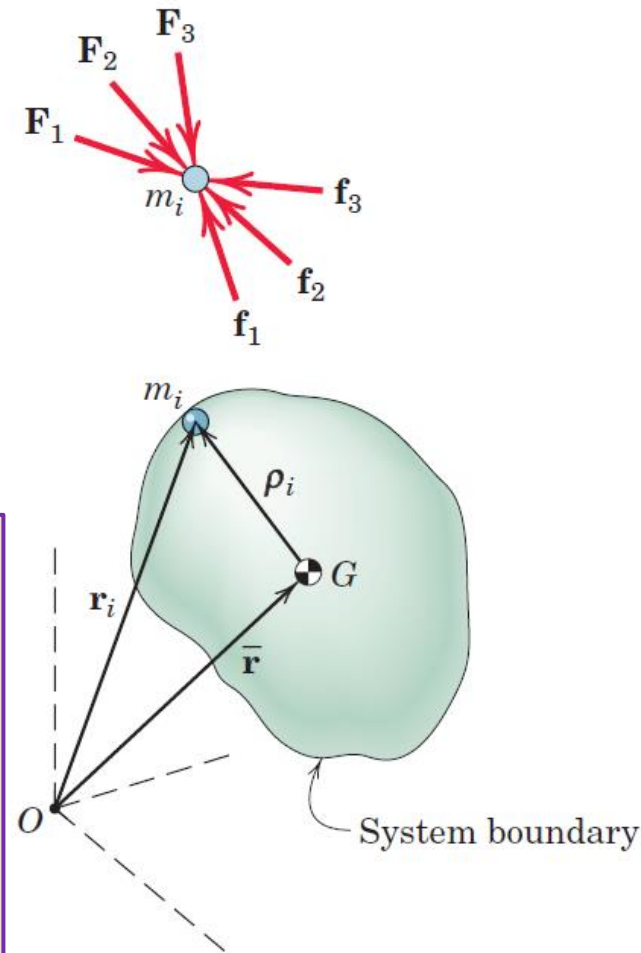
Generalized Newton's Second Law

Sum for all n mass particles

$$\sum F = m\bar{a}$$

$\bar{a}$ is the acceleration of the mathematical point which represents instantaneously the position of the mass center for the given n particles

- Acceleration vector have the same direction as the resultant external force $\sum F$
- $\sum F$ NOT necessarily passes through G



Work-Energy

Work energy equation for i^{th} particle

$$U_{1-2_i} = \Delta T_i$$

U_{1-2_i} is work done by external and internal forces.

Sum of work energy equation for all the particles

$$\sum_{i=1}^n U_{1-2_i} = \sum_{i=1}^n \Delta T_i$$

$\sum_{i=1}^n U_{1-2_i}$ includes work done by external forces and internal forces.

$\sum_{i=1}^n \Delta T_i$: Change in total kinetic energy of the system

Work-Energy

Sum of work energy equation for all the particles

$$U_{1-2}' = \Delta E = \Delta T + \Delta V$$

$$\Delta V = V_g + V_e$$

V_g : Gravitational potential energy

V_e : Elastic potential energy stored in spring

E : Total Mechanical Energy

$$U_{1-2}' + T_1 + V_1 = T_2 + V_2$$

Kinetic Energy

Sum for all n mass particles

$$T = \sum \frac{1}{2} m_i v_i^2 \quad \mathbf{v}_i = \bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_i$$

$\mathbf{v}_i$ is the velocity of i^{th} particle. $\bar{\mathbf{v}}$ is velocity of mass center G .

$\dot{\boldsymbol{\rho}}_i$ is the velocity of i^{th} particle with respect to translating frame of reference moving with mass center G .

$$v_i^2 = \mathbf{v}_i \cdot \mathbf{v}_i$$

$$T = \sum \frac{1}{2} m_i \mathbf{v}_i \cdot \mathbf{v}_i = \sum \frac{1}{2} m_i (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_i) \cdot (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_i)$$

$$T = \sum \frac{1}{2} m_i (\bar{\mathbf{v}} \cdot \bar{\mathbf{v}} + 2 \dot{\boldsymbol{\rho}}_i \cdot \bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_i \cdot \dot{\boldsymbol{\rho}}_i)$$

Kinetic Energy

$$T = \sum \frac{1}{2} m_i \mathbf{v}_i \cdot \mathbf{v}_i = \sum \frac{1}{2} m_i (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_i) \cdot (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_i)$$

$$T = \sum \frac{1}{2} (m_i \bar{\mathbf{v}} \cdot \bar{\mathbf{v}} + 2m_i \dot{\boldsymbol{\rho}}_i \cdot \bar{\mathbf{v}} + m_i \dot{\boldsymbol{\rho}}_i \cdot \dot{\boldsymbol{\rho}}_i)$$

$$\sum (m_i \boldsymbol{\rho}_i) = 0$$

$$\bar{\mathbf{v}} \cdot \frac{d}{dt} \left(\sum (m_i \boldsymbol{\rho}_i) \right) = 0$$

$$T = \sum \frac{1}{2} (m_i \bar{v}^2 + m_i |\dot{\boldsymbol{\rho}}_i|^2)$$

Kinetic Energy

$$T = \sum \frac{1}{2} m_i \mathbf{v}_i \cdot \mathbf{v}_i = \sum \frac{1}{2} m_i (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_i) \cdot (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_i)$$

$$T = \sum \frac{1}{2} (m_i \bar{\mathbf{v}} \cdot \bar{\mathbf{v}} + 2m_i \dot{\boldsymbol{\rho}}_i \cdot \bar{\mathbf{v}} + m_i \dot{\boldsymbol{\rho}}_i \cdot \dot{\boldsymbol{\rho}}_i)$$

$$T = \sum \frac{1}{2} (m_i \bar{v}^2 + m_i |\dot{\boldsymbol{\rho}}_i|^2) = \frac{1}{2} \left(m \bar{v}^2 + \sum m_i |\dot{\boldsymbol{\rho}}_i|^2 \right)$$

Total kinetic energy of a mass system equals the kinetic energy of mass-center translation of the system as a whole plus the kinetic energy due to motion of all particles relative to the mass center.

Linear Impulse-Momentum

Momentum of i^{th} particle of mass

$$\mathbf{G}_i = m_i \mathbf{v}_i$$

Linear momentum of the system: Vector sum of the linear momenta of all of its particles

$$\mathbf{G} = \sum m_i \mathbf{v}_i \qquad \mathbf{v}_i = \bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_i$$

$$\mathbf{G} = \sum m_i (\bar{\mathbf{v}} + \dot{\boldsymbol{\rho}}_i) = \sum m_i \bar{\mathbf{v}} + \sum m_i \dot{\boldsymbol{\rho}}_i = m \bar{\mathbf{v}} + 0$$

Linear momentum of any system of constant mass is the product of the mass and the velocity of its center of mass.

Linear Impulse-Momentum

$$\mathbf{G} = m\bar{\mathbf{v}}$$

$$\dot{\mathbf{G}} = m\dot{\bar{\mathbf{v}}} = m\mathbf{a} = \sum \mathbf{F}$$

- The resultant of the external forces on any mass system equals the time rate of change of the linear momentum of the system
- $\sum \mathbf{F}$, in general, does not pass through the mass center G .
- The equation does not apply to systems whose mass changes with time.

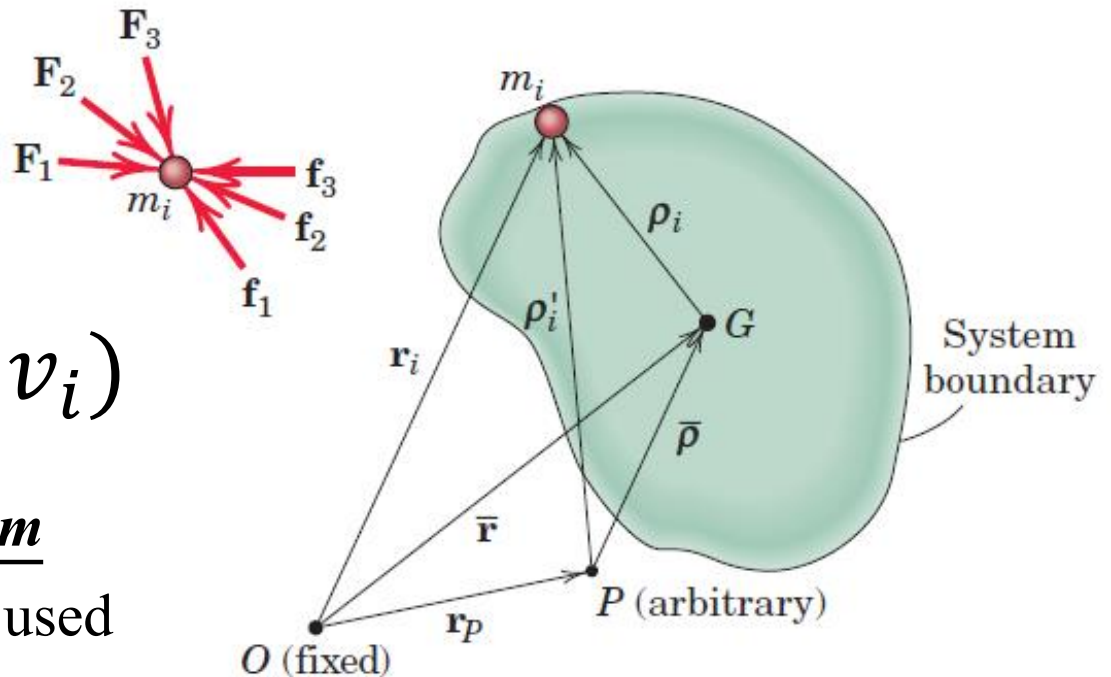
Angular Momentum about Fixed Point

The angular momentum of the mass system about the point O , fixed in the Newtonian reference system, is defined as the vector sum of the moments of the linear momenta about O of all particles of the system

$$H_O = \sum (r_i \times m_i v_i)$$

Absolute angular momentum

because absolute velocity is used



Angular Momentum about Fixed Point

$$\dot{H}_o = \sum (\dot{r}_i \times m_i v_i) + (r_i \times m_i \dot{v}_i)$$

$$\dot{H}_o = \sum (r_i \times m_i \dot{v}_i) = \sum (r_i \times F_i)$$

$$\dot{H}_o = \sum M_o$$

The moment sum $\sum M_o$ represents the moments of forces acting on the particles.

Angular Momentum about Fixed Point

$$\dot{H}_o = \sum (r_i \times m_i \dot{v}_i) = \sum (r_i \times F_i) = \sum M_o$$

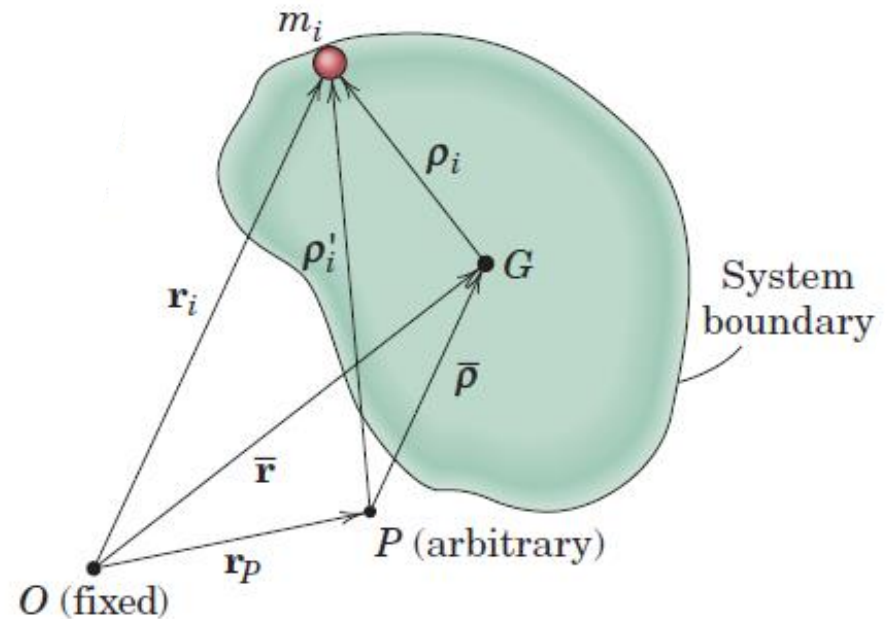
- The resultant vector moment about any fixed point of all the forces on any system of mass equals the time rate of change of angular momentum of the system about the fixed point

Angular Momentum about Mass Center, G

Sum of the moments of the linear momenta about G of all particles

$$H_G = \sum \rho_i \times m_i \dot{\mathbf{r}}_i$$

$$\dot{\mathbf{r}}_i = (\dot{\bar{\mathbf{r}}} + \dot{\rho}_i)$$



$$H_G = \sum \rho_i \times m_i (\dot{\bar{\mathbf{r}}} + \dot{\rho}_i)$$

$$H_G = \sum \rho_i \times m_i (\dot{\bar{\mathbf{r}}} + \dot{\rho}_i)$$

Angular Momentum about Mass Center, G

Sum of the moments of the linear momenta about G of all particles

$$H_G = \sum \boldsymbol{\rho}_i \times m_i \dot{\mathbf{r}}_i \quad \dot{\mathbf{r}}_i = (\dot{\mathbf{r}} + \dot{\boldsymbol{\rho}}_i)$$

$$H_G = \sum \boldsymbol{\rho}_i \times m_i (\dot{\mathbf{r}} + \dot{\boldsymbol{\rho}}_i)$$

$$H_G = \sum \boldsymbol{\rho}_i \times m_i \dot{\mathbf{r}} + \sum \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i$$

$$H_G = \sum -\dot{\mathbf{r}} \times m_i \boldsymbol{\rho}_i + \sum \boldsymbol{\rho}_i \times \frac{dm_i \boldsymbol{\rho}_i}{dt}$$

Angular Momentum about Mass Center, G

Sum of the moments of the linear momenta about G of all particles

$$H_G = -\dot{\mathbf{r}} \times \sum m_i \boldsymbol{\rho}_i + \sum \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i$$

$$H_G = 0 + \sum \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i$$

$$H_G = \sum \boldsymbol{\rho}_i \times m_i \dot{\boldsymbol{\rho}}_i$$

Relative angular momentum because the relative velocity is used

Angular momentum about point G is equal to relative angular momentum about point G .

Angular Momentum about Mass Center, G

Sum of the moments of the linear momenta about G of all particles

$$\mathbf{H}_G = \sum \boldsymbol{\rho}_i \times m_i \dot{\mathbf{r}}_i$$

$$\dot{\mathbf{r}}_i = (\dot{\mathbf{r}} + \dot{\boldsymbol{\rho}}_i) \quad \mathbf{H}_G = \sum \boldsymbol{\rho}_i \times m_i (\dot{\mathbf{r}} + \dot{\boldsymbol{\rho}}_i)$$

$$\dot{\mathbf{H}}_G = \sum \dot{\boldsymbol{\rho}}_i \times m_i \dot{\mathbf{r}}_i + \sum \boldsymbol{\rho}_i \times m_i \ddot{\mathbf{r}}_i$$

$$\dot{\mathbf{H}}_G = \sum \dot{\boldsymbol{\rho}}_i \times m_i (\dot{\mathbf{r}} + \dot{\boldsymbol{\rho}}_i) + \sum \boldsymbol{\rho}_i \times m_i \ddot{\mathbf{r}}_i$$

Angular Momentum about Mass Center, G

Sum of the moments of the linear momenta about G of all particles

$$\dot{\mathbf{H}}_G = \sum \dot{\boldsymbol{\rho}}_i \times m_i (\dot{\mathbf{r}} + \dot{\boldsymbol{\rho}}_i) + \sum \boldsymbol{\rho}_i \times m_i \ddot{\mathbf{r}}_i$$

$$\dot{\mathbf{H}}_G = \sum \dot{\boldsymbol{\rho}}_i \times m_i \dot{\mathbf{r}} + \sum \dot{\boldsymbol{\rho}}_i \times m_i \dot{\boldsymbol{\rho}}_i + \sum \boldsymbol{\rho}_i \times m_i \ddot{\mathbf{r}}_i$$

$$\dot{\mathbf{H}}_G = -\dot{\mathbf{r}} \times \frac{d \sum (m_i \rho_i)}{dt} + 0 + \sum \boldsymbol{\rho}_i \times m_i \ddot{\mathbf{r}}_i$$

$$\dot{\mathbf{H}}_G = 0 + 0 + \sum \boldsymbol{\rho}_i \times m_i \ddot{\mathbf{r}}_i$$

Angular Momentum about Mass Center, G

Sum of the moments of the linear momenta about G of all particles

$$\dot{\mathbf{H}}_G = \sum \boldsymbol{\rho}_i \times m_i \ddot{\mathbf{r}}_i = \sum \boldsymbol{\rho}_i \times (\mathbf{F}_i + \mathbf{f}_i) = \sum \mathbf{M}_G$$

$$\dot{\mathbf{H}}_G = \sum \boldsymbol{\rho}_i \times (\mathbf{F}_i + \mathbf{f}_i) = \sum \mathbf{M}_G$$

$$\dot{\mathbf{H}}_G = \sum \mathbf{M}_G$$

Rate of change of angular momentum about point G is sum of all moments about point G

Angular Momentum about Arbitrary Point, P

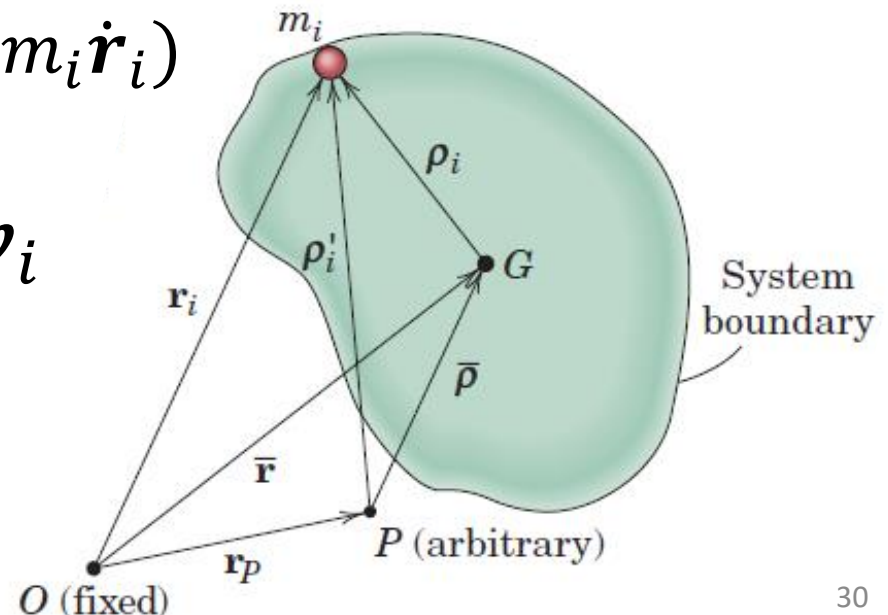
Sum of the moments of the linear momenta about P (which may have acceleration) of all particles

$$\mathbf{H}_P = \sum \boldsymbol{\rho}'_i \times m_i \dot{\mathbf{r}}_i = \sum (\bar{\boldsymbol{\rho}} + \boldsymbol{\rho}_i) \times m_i \dot{\mathbf{r}}_i$$

$$\mathbf{H}_P = \sum (\bar{\boldsymbol{\rho}} \times m_i \dot{\mathbf{r}}_i + \boldsymbol{\rho}_i \times m_i \dot{\mathbf{r}}_i)$$

$$\mathbf{H}_P = \mathbf{H}_G + \bar{\boldsymbol{\rho}} \times \sum m_i \mathbf{v}_i$$

$$\mathbf{H}_P = \mathbf{H}_G + \bar{\boldsymbol{\rho}} \times m \bar{\mathbf{v}}$$

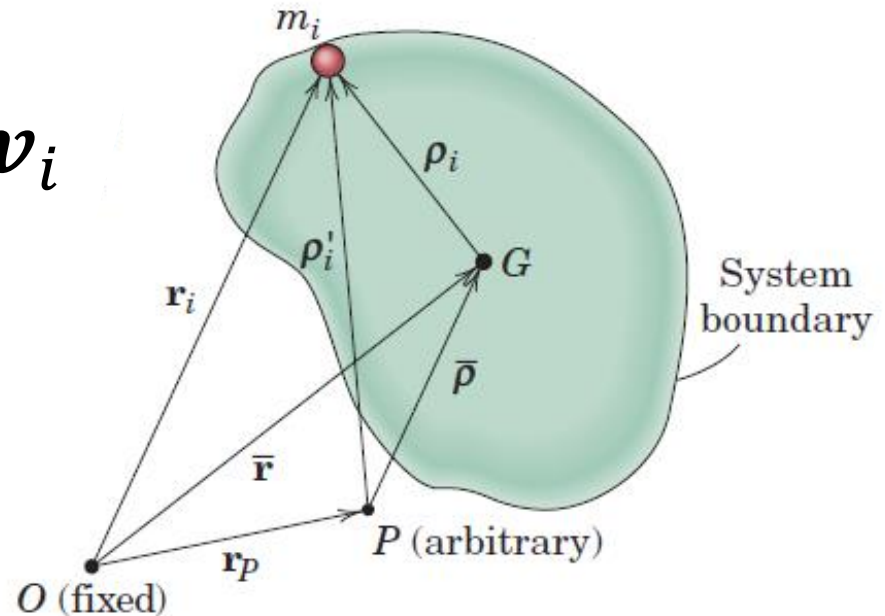


Angular Momentum about Arbitrary Point, P

Sum of the moments of the linear momenta about P of all particles

$$\mathbf{H}_P = \mathbf{H}_G + \bar{\boldsymbol{\rho}} \times \sum m_i \mathbf{v}_i$$

$$\mathbf{H}_P = \mathbf{H}_G + \bar{\boldsymbol{\rho}} \times m \bar{\mathbf{v}}$$

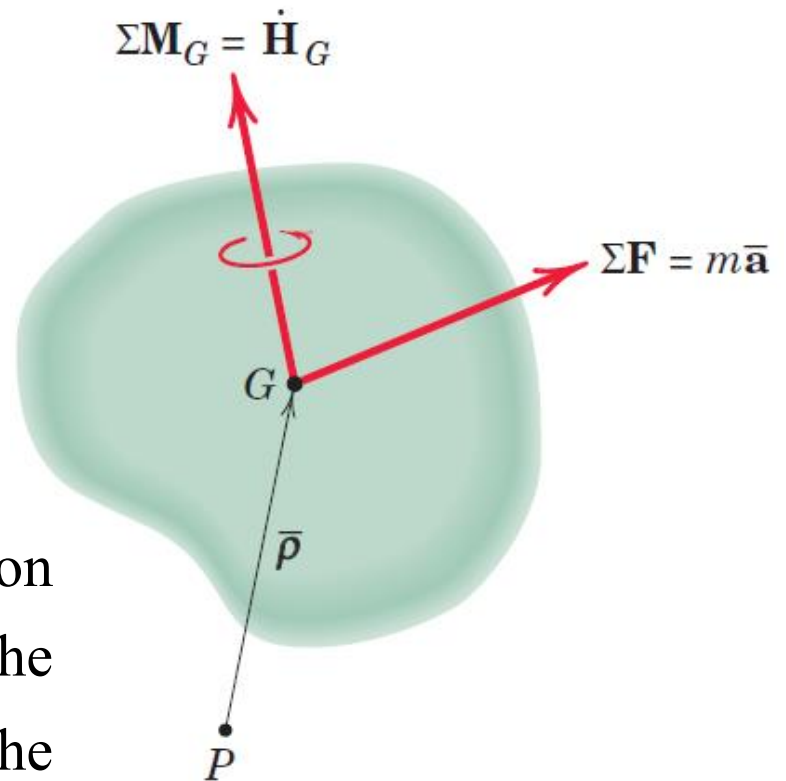


Absolute angular momentum about any point P equals the angular momentum about G plus the moment about P of the linear momentum of the system considered concentrated at G

Angular Momentum about Arbitrary Point, P

Representing a force system by a resultant force through any point, such as G , and a corresponding couple

Resultant of the external forces acting on the system expressed in terms of the resultant force $\sum \mathbf{F}$ through G and the corresponding couple $\sum M_G$



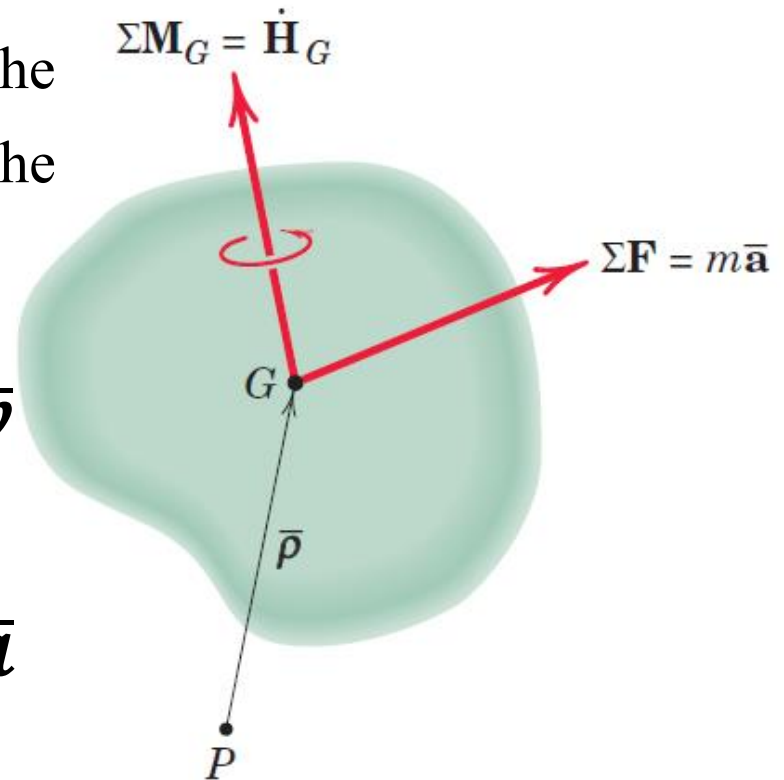
Angular Momentum about Arbitrary Point, P

Resultant of the external forces acting on the system expressed in terms of the resultant force $\Sigma \mathbf{F}$ through G and the corresponding couple $\Sigma \mathbf{M}_G$

$$\Sigma \mathbf{M}_P = \Sigma \mathbf{M}_G + \bar{\boldsymbol{\rho}} \times m \dot{\mathbf{v}}$$

$$\Sigma \mathbf{M}_P = \Sigma \mathbf{M}_G + \bar{\boldsymbol{\rho}} \times m \bar{\mathbf{a}}$$

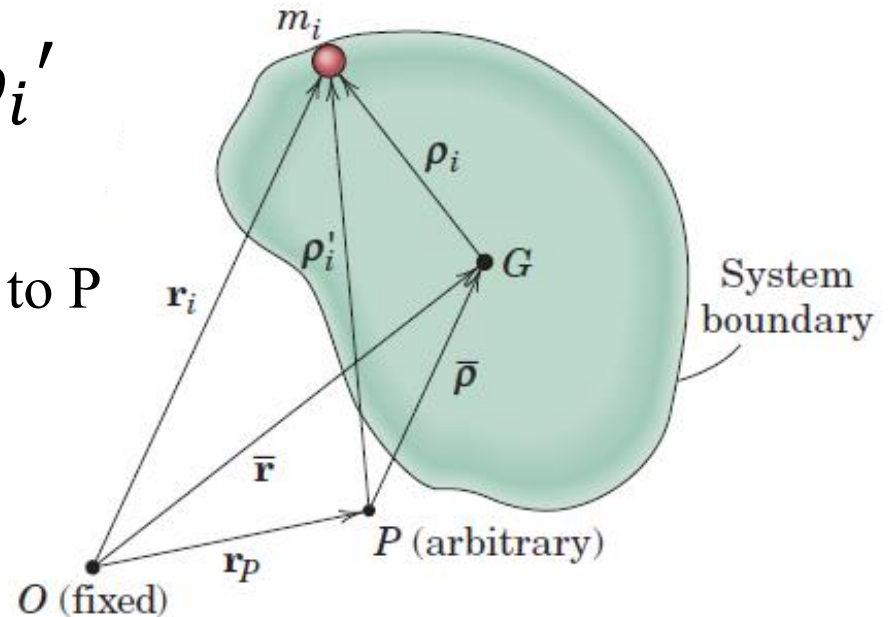
$$\Sigma \mathbf{M}_P = \dot{\mathbf{H}}_G + \bar{\boldsymbol{\rho}} \times m \bar{\mathbf{a}}$$



Relative Angular Momentum about Arbitrary Point, P

$$(\mathbf{H}_P)_{rel} = \sum \rho_i' \times m_i \dot{\rho}_i'$$

$\dot{\rho}_i'$ is velocity of m_i with respect to P



$$(\mathbf{H}_P)_{rel} = \sum (\bar{\rho} + \rho_i) \times m_i (\dot{\bar{\rho}} + \dot{\rho}_i)$$

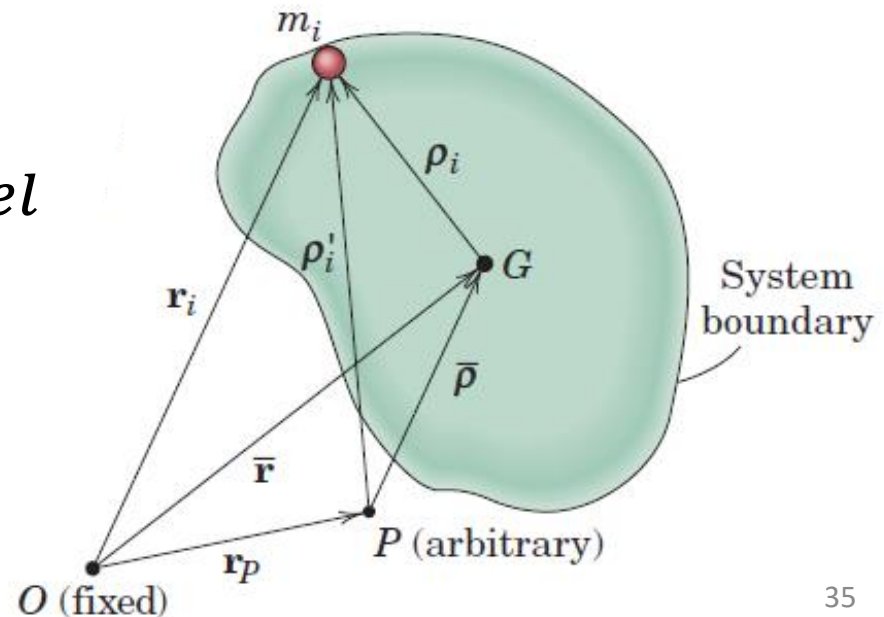
Relative Angular Momentum about Arbitrary Point, P

$$(\mathbf{H}_P)_{rel} = \sum (\bar{\rho} + \rho_i) \times m_i(\dot{\bar{\rho}} + \dot{\rho}_i)$$

$$(\mathbf{H}_P)_{rel} = \sum (\bar{\rho} \times m_i \dot{\bar{\rho}} + \bar{\rho} \times m_i \dot{\rho}_i + \rho_i \times m_i \dot{\bar{\rho}} + \rho_i \times m_i \dot{\rho}_i)$$

$$\sum \bar{\rho} \times m_i \dot{\bar{\rho}} = \bar{\rho} \times m v_{rel}$$

$$\sum \bar{\rho} \times \frac{d(m_i \rho_i)}{dt} = 0$$



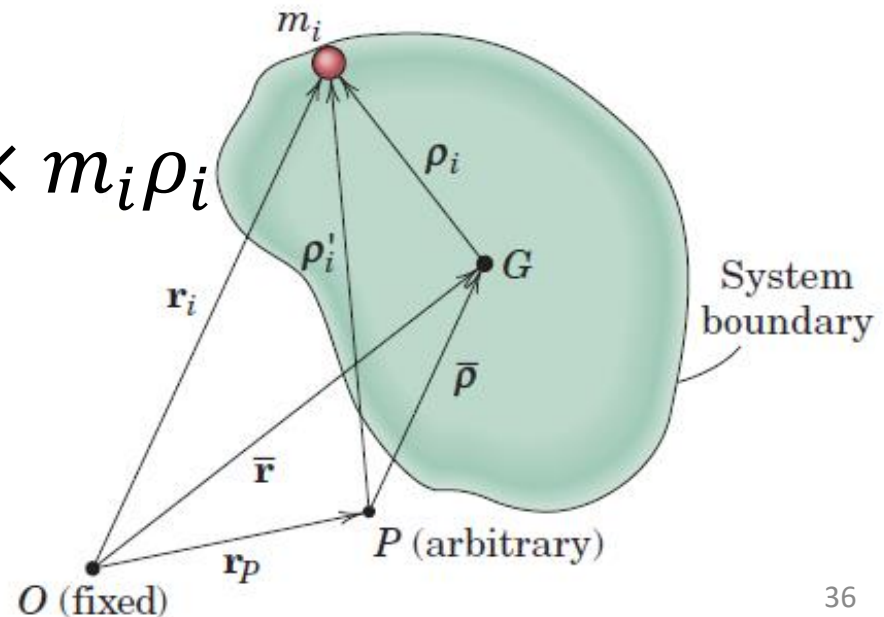
Relative Angular Momentum about Arbitrary Point, P

$$(\mathbf{H}_P)_{rel} = \sum (\bar{\rho} + \rho_i) \times m_i (\dot{\bar{\rho}} + \dot{\rho}_i)$$

$$(\mathbf{H}_P)_{rel} = \sum (\bar{\rho} \times m_i \dot{\bar{\rho}} + \bar{\rho} \times m_i \dot{\rho}_i + \rho_i \times m_i \dot{\bar{\rho}} + \rho_i \times m_i \dot{\rho}_i)$$

$$\sum \rho_i \times m_i \dot{\bar{\rho}} = \sum -\dot{\bar{\rho}} \times m_i \rho_i$$

$$-\dot{\bar{\rho}} \times \sum m_i \rho_i = 0$$



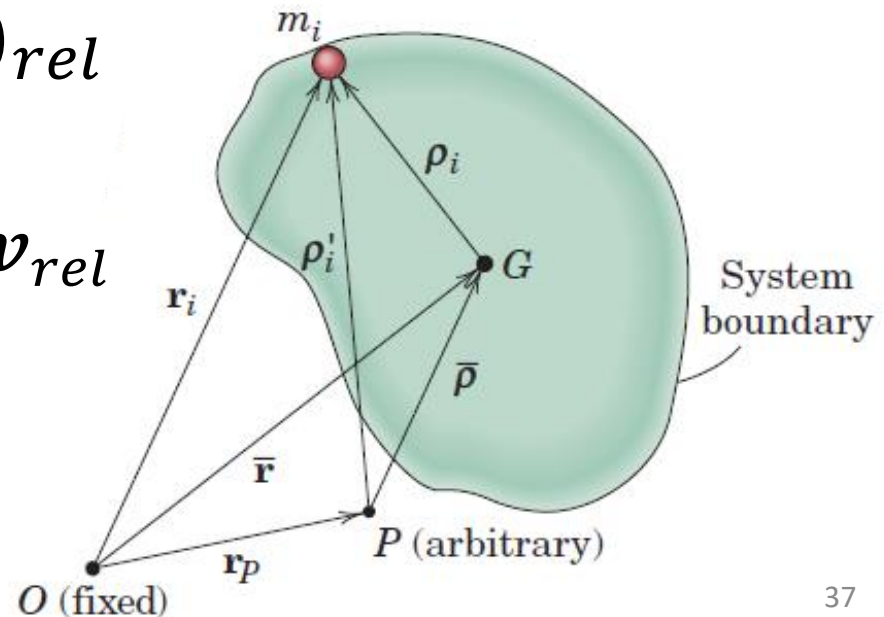
Relative Angular Momentum about Arbitrary Point, P

$$(\mathbf{H}_P)_{rel} = \sum (\bar{\rho} + \rho_i) \times m_i (\dot{\bar{\rho}} + \dot{\rho}_i)$$

$$(\mathbf{H}_P)_{rel} = \sum (\bar{\rho} \times m_i \dot{\bar{\rho}} + \bar{\rho} \times m_i \dot{\rho}_i + \rho_i \times m_i \dot{\bar{\rho}} + \rho_i \times m_i \dot{\rho}_i)$$

$$\sum \rho_i \times m_i \dot{\rho}_i = (H_G)_{rel}$$

$$(\mathbf{H}_P)_{rel} = (\mathbf{H}_G)_{rel} + \bar{\rho} \times m \mathbf{v}_{rel}$$



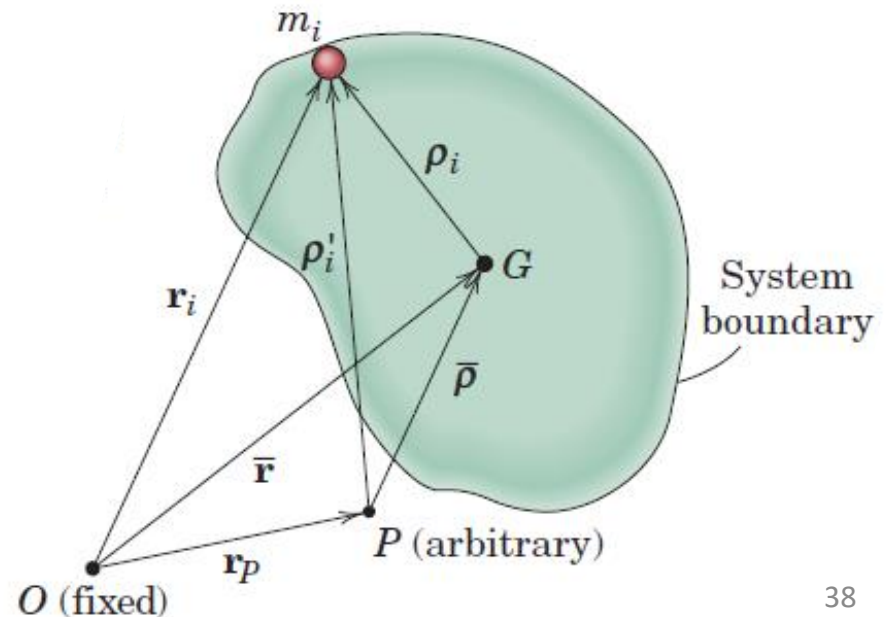
Relative Angular Momentum about Arbitrary Point, P

$$(\mathbf{H}_P)_{rel} = \sum (\boldsymbol{\rho}_i') \times m_i(\dot{\boldsymbol{\rho}}_i')$$

$$(\dot{\mathbf{H}}_P)_{rel} = \sum (\dot{\boldsymbol{\rho}}_i') \times m_i(\dot{\boldsymbol{\rho}}_i') + (\boldsymbol{\rho}_i') \times m_i(\ddot{\boldsymbol{\rho}}_i')$$

$$\mathbf{r}_i = \mathbf{r}_p + (\boldsymbol{\rho}_i')$$

$$\ddot{\mathbf{r}}_i = \ddot{\mathbf{r}}_p + (\ddot{\boldsymbol{\rho}}_i')$$

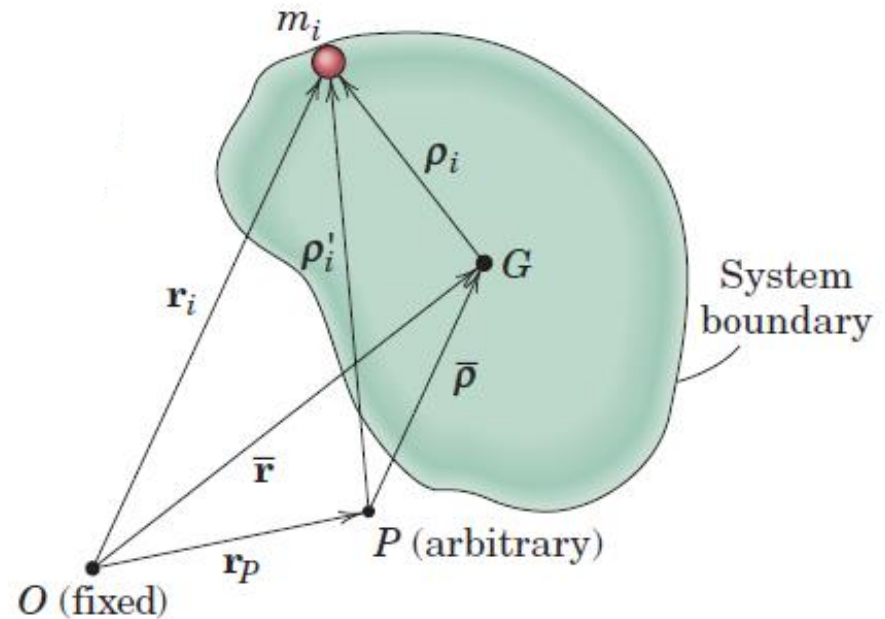


Relative Angular Momentum about Arbitrary Point, P

$$(\dot{\mathbf{H}}_P)_{rel} = \sum (\dot{\rho}_i') \times m_i(\dot{\rho}_i') + (\rho_i') \times m_i(\ddot{\rho}_i')$$

$$\mathbf{r}_i = \mathbf{r}_p + (\rho_i')$$

$$(\ddot{\rho}_i') = \ddot{\mathbf{r}}_i - \ddot{\mathbf{r}}_p$$



$$(\dot{\mathbf{H}}_P)_{rel} = \sum (\rho_i') \times m_i(\ddot{\mathbf{r}}_i - \ddot{\mathbf{r}}_p)$$

Relative Angular Momentum about Arbitrary Point, P

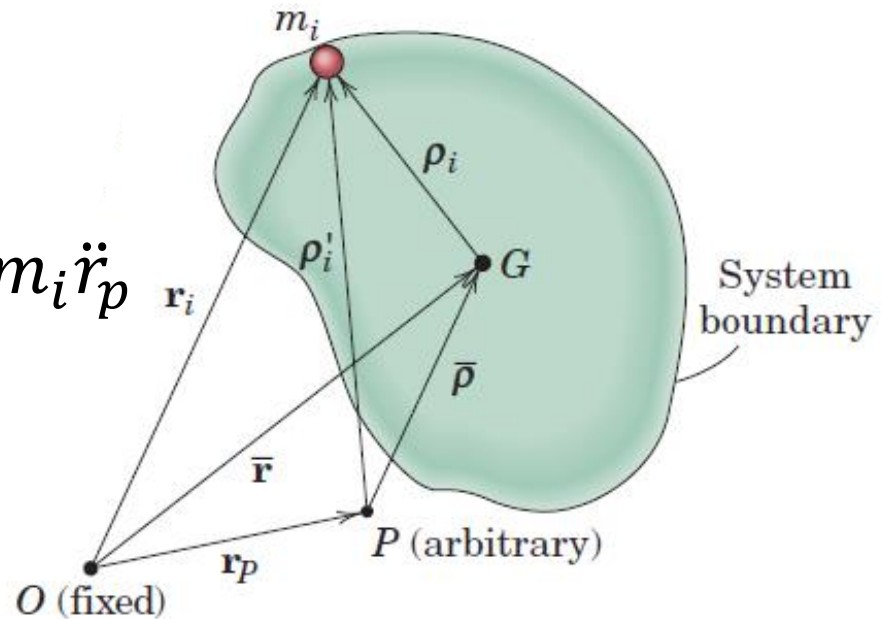
$$(\dot{\mathbf{H}}_P)_{rel} = \sum (\rho_i') \times m_i (\ddot{\mathbf{r}}_i - \ddot{\mathbf{r}}_p)$$

$$(\dot{\mathbf{H}}_P)_{rel}$$

$$= \sum (\rho_i') \times m_i \ddot{\mathbf{r}}_i - (\rho_i') \times m_i \ddot{\mathbf{r}}_p$$

$$(\dot{\mathbf{H}}_P)_{rel}$$

$$= \sum \mathbf{M}_P + \sum -(\rho_i') \times m_i \ddot{\mathbf{r}}_p$$



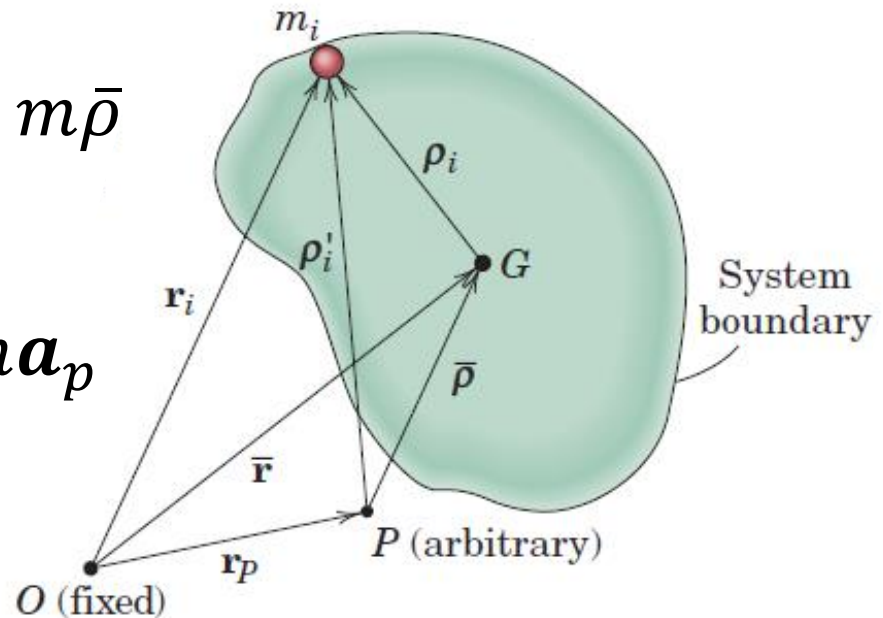
Relative Angular Momentum about Arbitrary Point, P

$$(\dot{\mathbf{H}}_P)_{rel} = \sum \mathbf{M}_P + \sum \ddot{\mathbf{r}}_p \times m_i (\boldsymbol{\rho}_i')$$

$$(\dot{\mathbf{H}}_P)_{rel} = \sum \mathbf{M}_P + \mathbf{a}_p \times \sum m_i (\boldsymbol{\rho}_i')$$

$$(\dot{\mathbf{H}}_P)_{rel} = \sum \mathbf{M}_P + \mathbf{a}_p \times m \bar{\boldsymbol{\rho}}$$

$$\sum \mathbf{M}_P = (\dot{\mathbf{H}}_P)_{rel} + \bar{\boldsymbol{\rho}} \times m \mathbf{a}_p$$



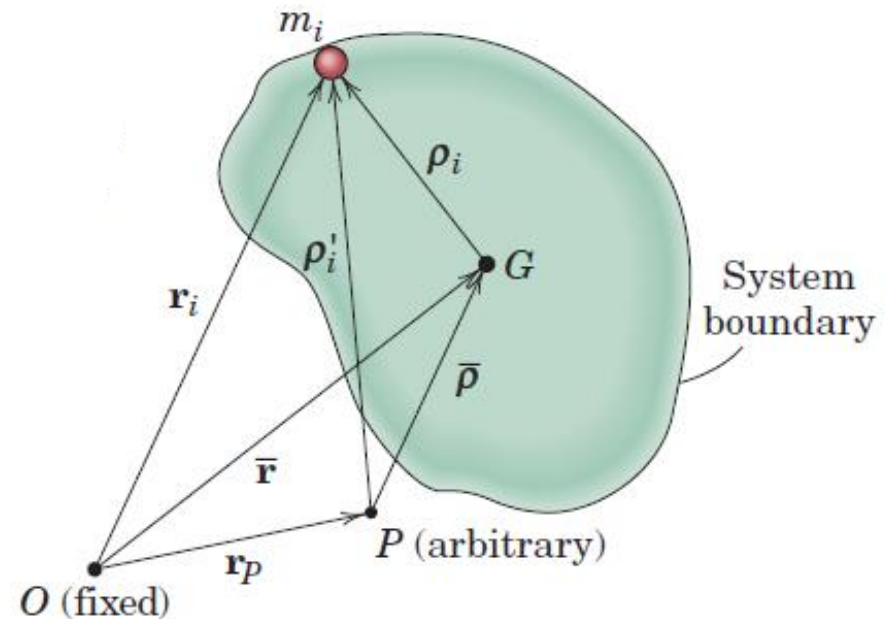
Relative Angular Momentum about Arbitrary Point, P

$$\sum \mathbf{M}_P = (\dot{\mathbf{H}}_P)_{rel} + \bar{\rho} \times m \mathbf{a}_p$$

If

- (a) $\bar{\rho} = 0$
- (b) $\mathbf{a}_p = 0$
- (c) $\bar{\rho}$ and $\mathbf{a}_p$ are parallel

$$\sum \mathbf{M}_P = (\dot{\mathbf{H}}_P)_{rel}$$



Law of Conservation of Dynamical Energy

- A mass system is said to be *conservative* if it does not lose energy by virtue of internal friction forces which do negative work or by virtue of inelastic members which dissipate energy upon cycling.
- If no work is done on a conservative system during an interval of motion by external forces (other than gravity or other potential forces), then none of the energy of the system is lost.

$$U'_{1-2} = 0$$

$$\Delta T + \Delta V = 0$$

$$T_1 + V_1 = T_2 + V_2$$

This law holds only in the ideal case where internal kinetic friction is sufficiently small to be neglected

Principle of Conservation of Linear Momentum

Resultant external force $\Sigma \mathbf{F}$ acting on a conservative or nonconservative mass system is zero,

$$\dot{\mathbf{G}} = \sum \mathbf{F} = 0$$

$$\mathbf{G}_1 = \mathbf{G}_2$$

In the absence of an external impulse, the linear momentum of a system remains unchanged

Principle of Conservation of Angular Momentum

Resultant moment about a fixed point O or about the mass center G of all forces on any mass system is zero,

$$\dot{\mathbf{H}}_O = \sum \mathbf{M}_O = 0 \qquad \dot{\mathbf{H}}_G = \sum \mathbf{M}_G = 0$$

$$\mathbf{H}_{O1} = \mathbf{H}_{O2}$$

$$\mathbf{H}_{G1} = \mathbf{H}_{G2}$$

There is no angular impulse about a fixed point (or about the mass center), the angular momentum of the system about the fixed point (or about the mass center) remains unchanged.

Application in a translating frame of reference

- The basic laws of Newtonian mechanics hold for measurements made relative to a set of axes which translate with a constant velocity.
- Thus, all equations, mentioned in previous slides, are valid provided all quantities are expressed relative to the translating axes moving with constant velocity.

Steady Mass Flow

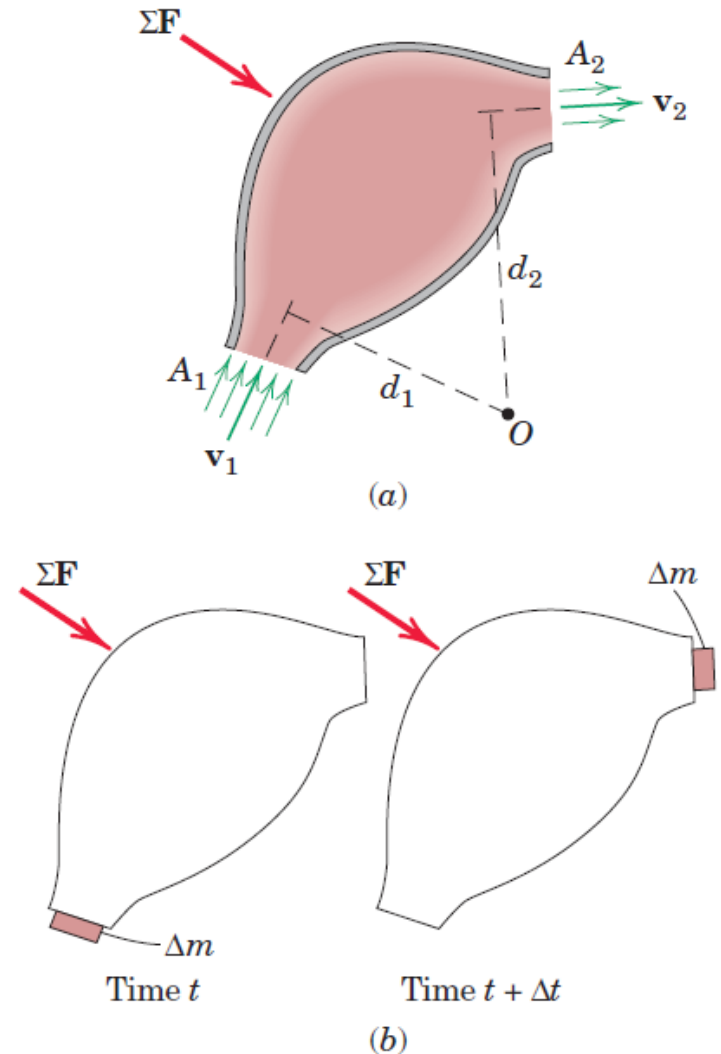
- **Steady Flow Condition:** Rate at which mass enters a given volume equals the rate at which mass leaves the same volume
- The volume may be enclosed by a **rigid container**, fixed or moving, such as the **nozzle of a jet aircraft or rocket**, the **space between blades in a gas turbine**, the **volume within the casing of a centrifugal pump**, the **volume within the bend of a pipe through which a fluid is flowing at a steady rate**
- Design of such fluid machines depends on the analysis of the forces and moments associated with the corresponding momentum changes of the flowing mass

Steady Mass Flow through a Rigid Container

- If ρ_1 and ρ_2 are the respective densities of the two streams, conservation of mass requires that

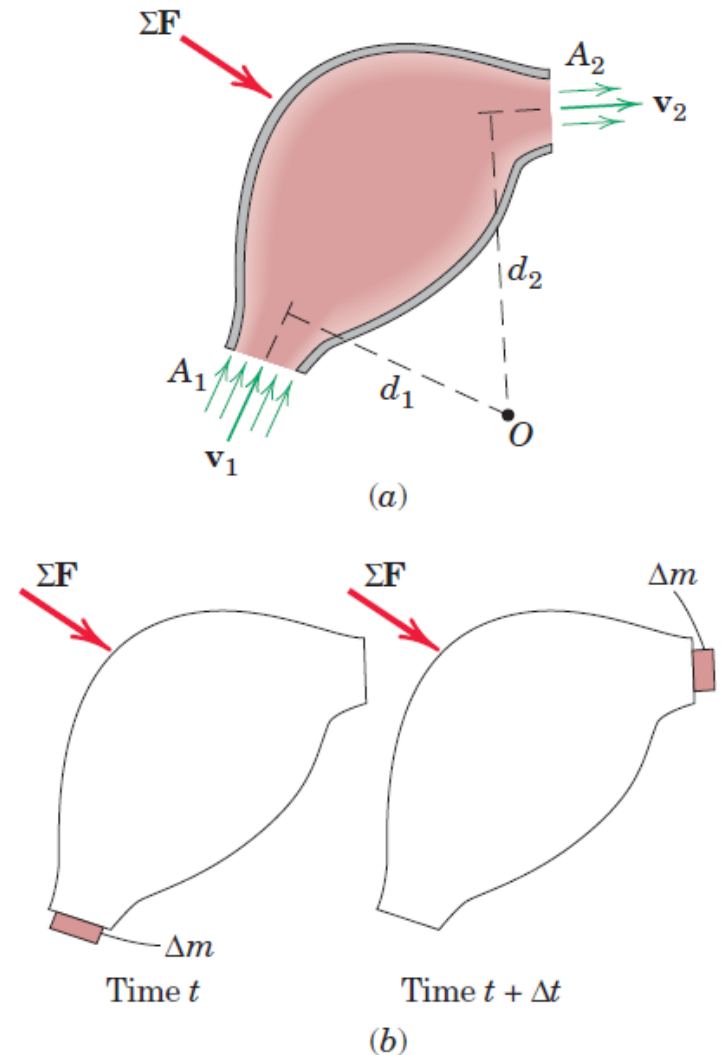
$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2 = m'$$

- Two approaches to isolate the system
1. Isolate the mass of fluid within the container (if the forces between the container and the fluid were to be described)
 2. Entire container and the fluid within it (when the forces external to the container are desired)



Steady Mass Flow through a Rigid Container

- Account for all forces applied externally to this system, ΣF
- Forces exerted on the container at points of its attachment to other structures, including attachments at A_1 and A_2 , if present
- Forces acting on the fluid within the container at A_1 and A_2 due to any static pressure which may exist in the fluid at these positions
- Weight of the fluid and structure if appreciable

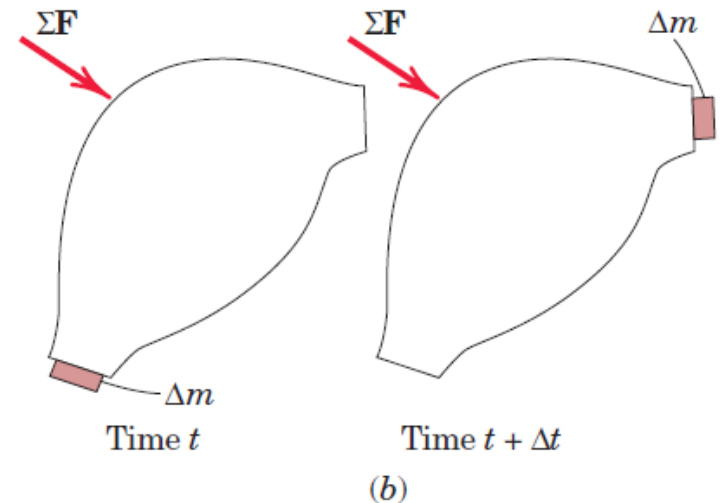
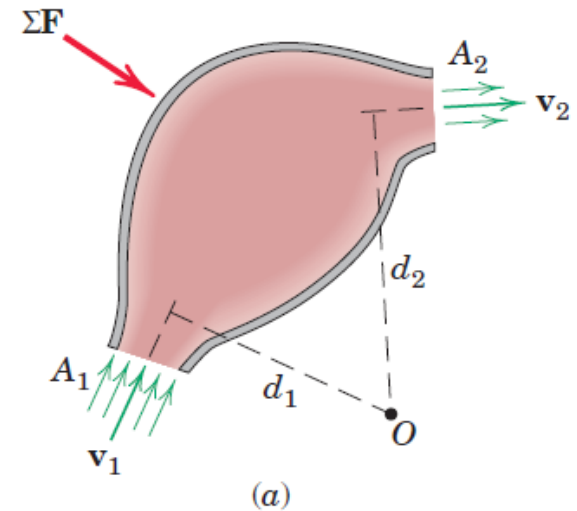


Steady Mass Flow through a Rigid Container

- All forces applied externally to this system, ΣF

$$\Sigma F = \dot{G}$$

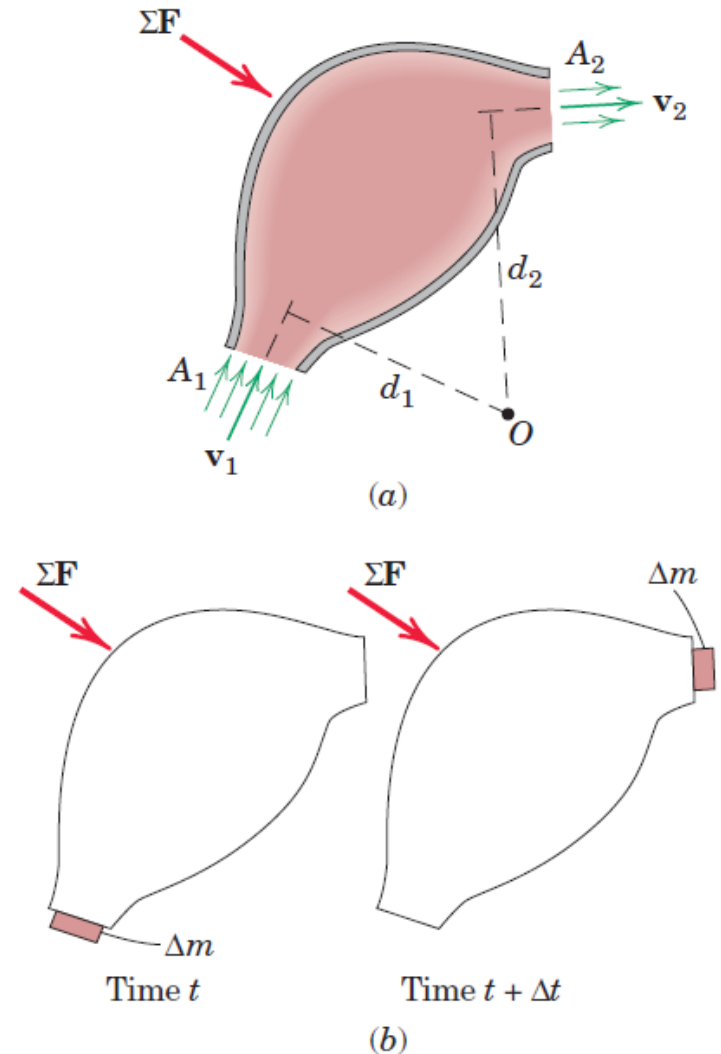
- The resultant of all of these external forces must equal the time rate of change of the linear momentum of the isolated system



Steady Mass Flow through a Rigid Container

Incremental Analysis

- The system at time t when the system mass is that of the container, the mass within it, and an increment Δm about to enter during time t . At time $t + \Delta t$ the same total mass is that of the container, the mass within it, and an equal increment Δm which leaves the container in time Δt .
- The linear momentum of the container and mass within it between the two sections A_1 and A_2 remains unchanged during Δt so that the change in momentum of the system in time Δt



Steady Mass Flow through a Rigid Container

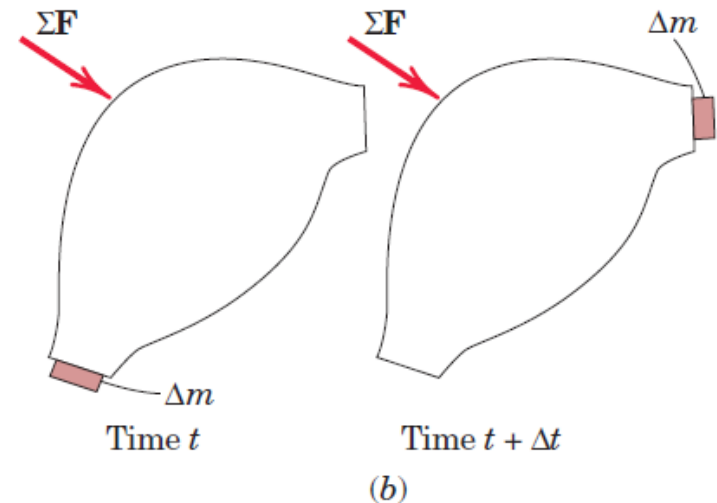
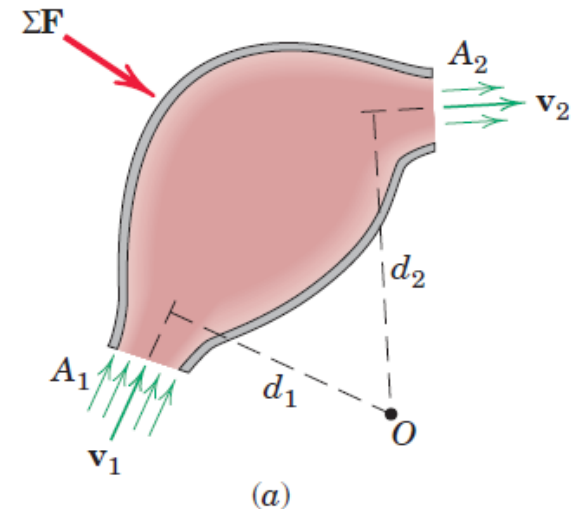
Incremental Analysis

- The change in linear momentum of the system in time Δt

$$\begin{aligned}\Delta G &= (\Delta m)v_2 - (\Delta m)v_1 \\ &= \Delta m(v_2 - v_1)\end{aligned}$$

$$m' = \lim_{\Delta t \rightarrow 0} \left(\frac{\Delta m}{\Delta t} \right) = \frac{dm}{dt}$$

$$\sum F = m' \Delta v = \dot{G}$$



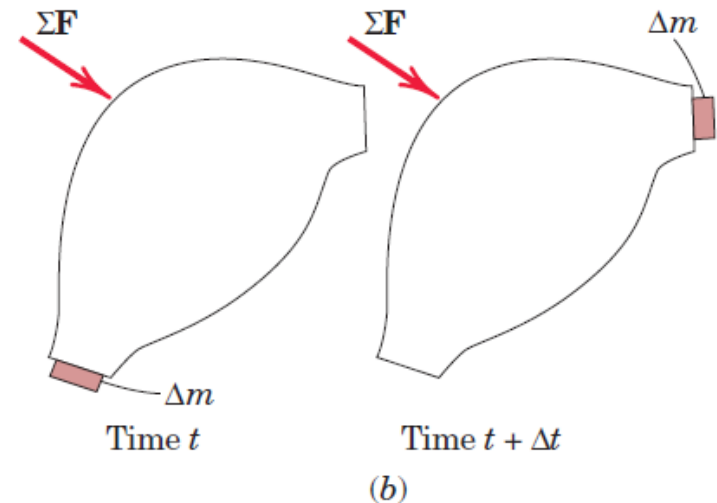
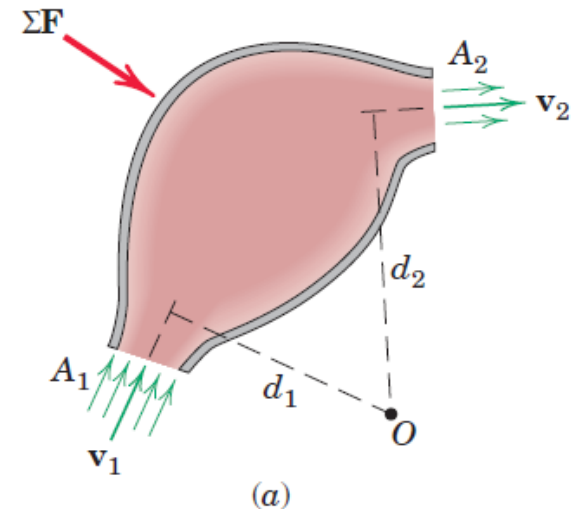
Steady Mass Flow through a Rigid Container

Incremental Analysis

- The change in linear momentum of the system in time Δt

$$\sum F = m' \Delta v = \dot{G}$$

Relation between the resultant force on a steady-flow system and the corresponding mass flow rate and vector velocity increment



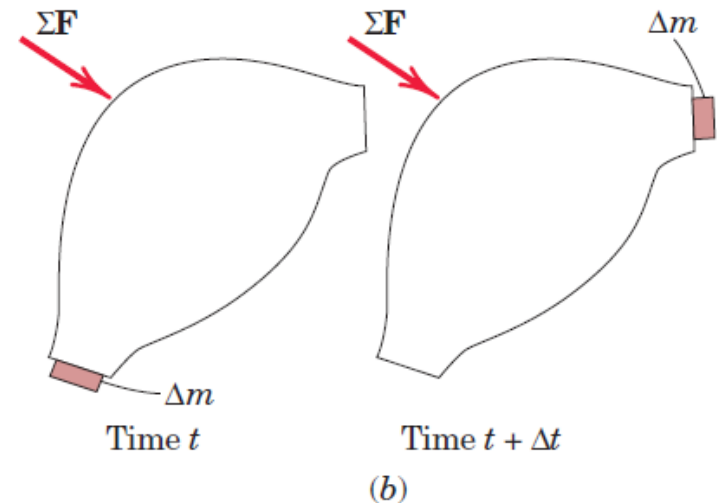
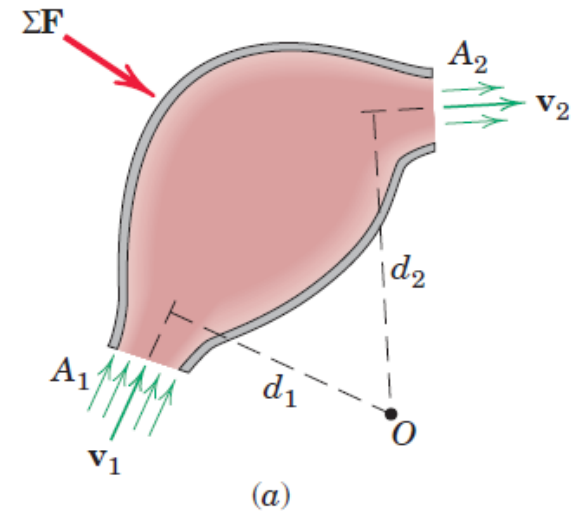
Steady Mass Flow through a Rigid Container

Incremental Analysis

- The change in linear momentum of the system in time Δt

$$\sum F = m' \Delta v = \dot{G}$$

Time rate of change of linear momentum is the vector difference between the rate at which linear momentum leaves the system and the rate at which linear momentum enters the system

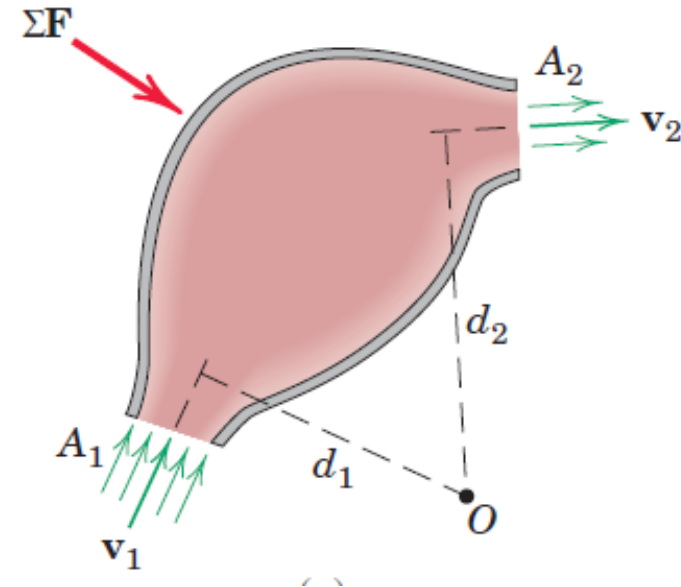


Angular Momentum in Steady Mass Flow System

- The resultant moment of all external forces about some fixed point O (which can be on or off the system) equals the time rate of change of angular momentum of the system about O

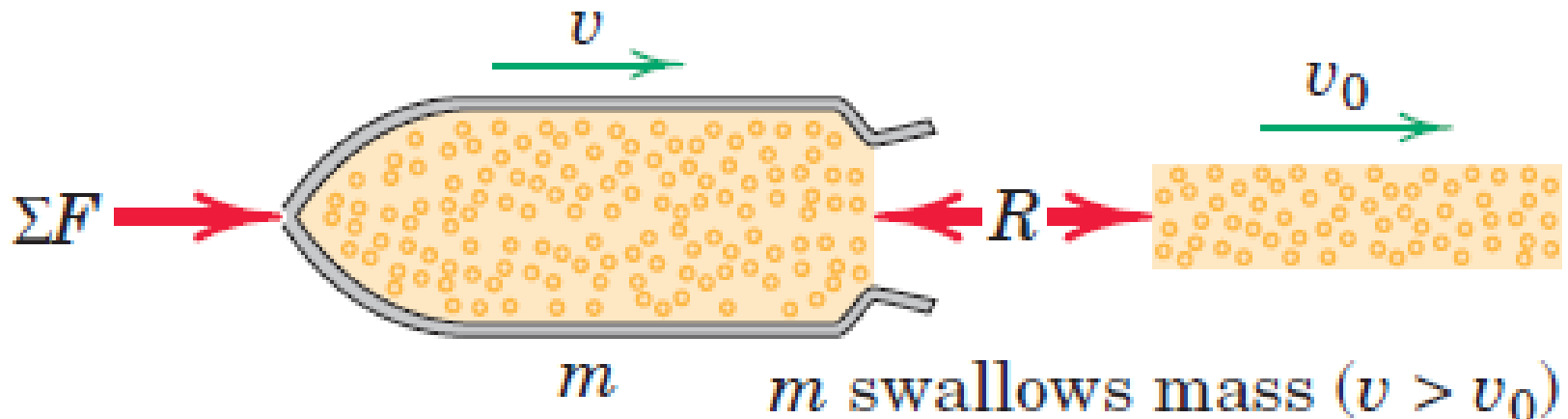
$$\sum M_0 = m'(v_2 d_2 - v_1 d_1)$$

$$\sum \mathbf{M}_0 = m'(\mathbf{d}_2 \times \mathbf{v}_2 - \mathbf{d}_1 \times \mathbf{v}_1)$$



Variable Mass Flow

- At any instant,
 Mass of the body, m
 Velocity at any instant, v
- Stream of matter is assumed to be moving in the same direction as m with a constant velocity $v_0 < v$
- Force exerted by m on particles of the stream to accelerate it from velocity v_0 to a higher velocity v , is
 $R = \dot{m}(v - v_0)$ where $\dot{m}$ is rate of increase of mass m



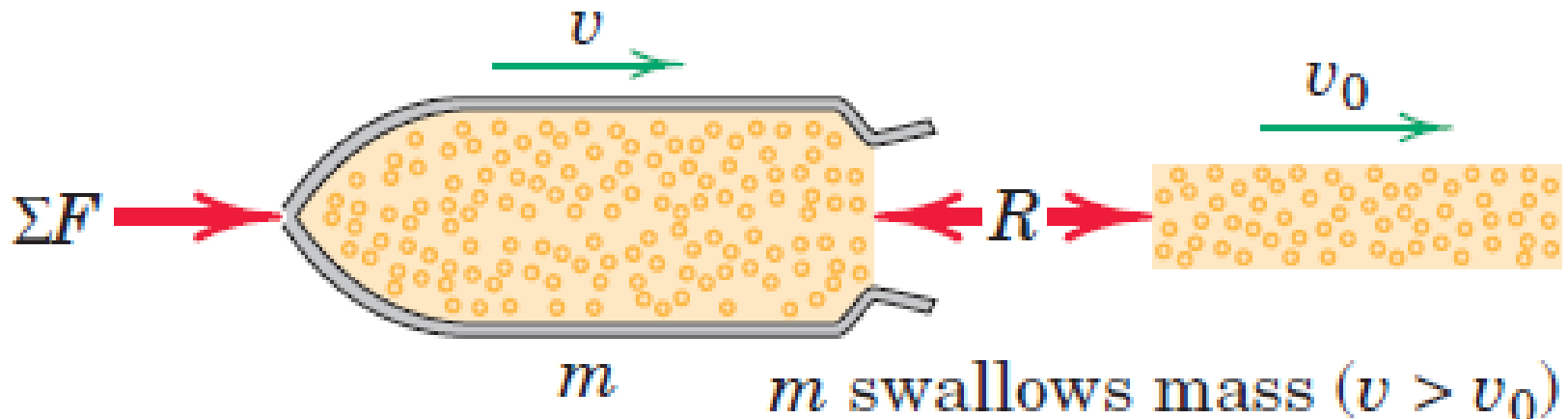
Variable Mass Flow

- Equation of motion for mass, m

$$\sum F - R = m\dot{v}$$

$$R = \dot{m}(v - v_0)$$

$$\sum F = m\dot{v} + \dot{m}(v - v_0)$$

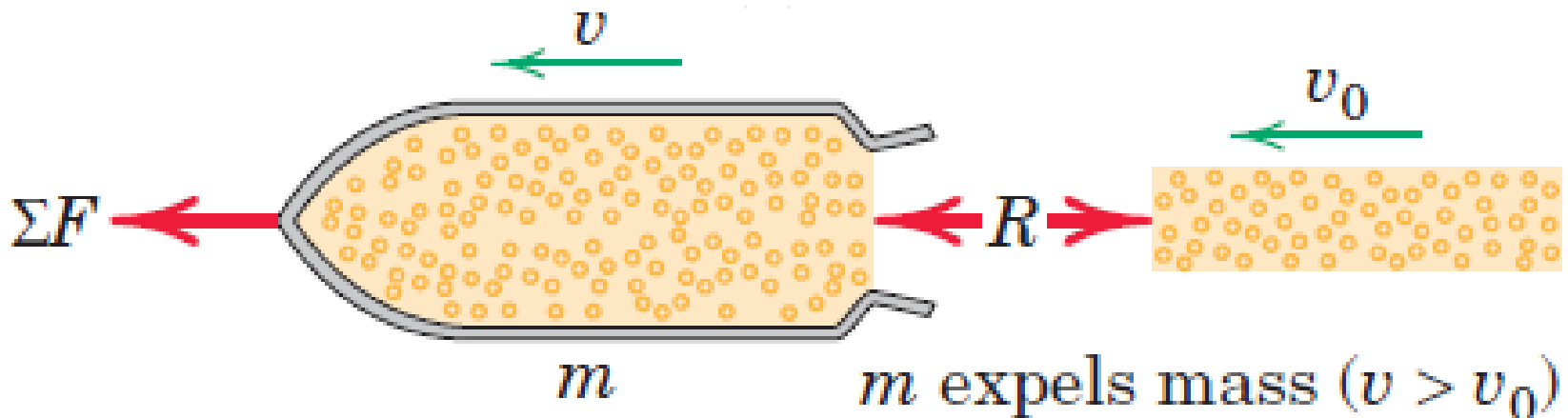


Variable Mass Flow

- Stream of matter is leaving mass m with a constant velocity $v_0 < v$
- Force exerted by m on particles of the stream to decelerate it from velocity v to a lower velocity v_0 , is

$R = -\dot{m}(-v_0 - (-v)) = -\dot{m}(v - v_0)$ where $-\dot{m}$ is rate of decrease of mass m .

$$\sum F + R = m\dot{v}$$



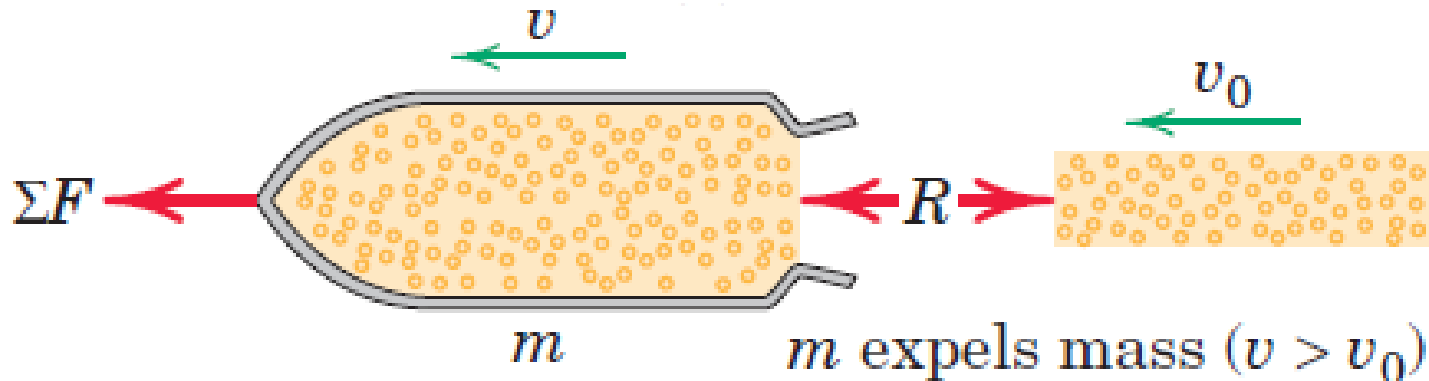
Variable Mass Flow

- Force exerted by m on particles of the stream to decelerate it from velocity v to a lower velocity v_0 , is

$R = -\dot{m}(-v_0 - (-v)) = -\dot{m}(v - v_0)$ where $-\dot{m}$ is rate of decrease of mass m .

$$\sum F + R = m\dot{v}$$

$$\sum F = m\dot{v} + \dot{m}(v - v_0) = m\dot{v} + \dot{m}u$$

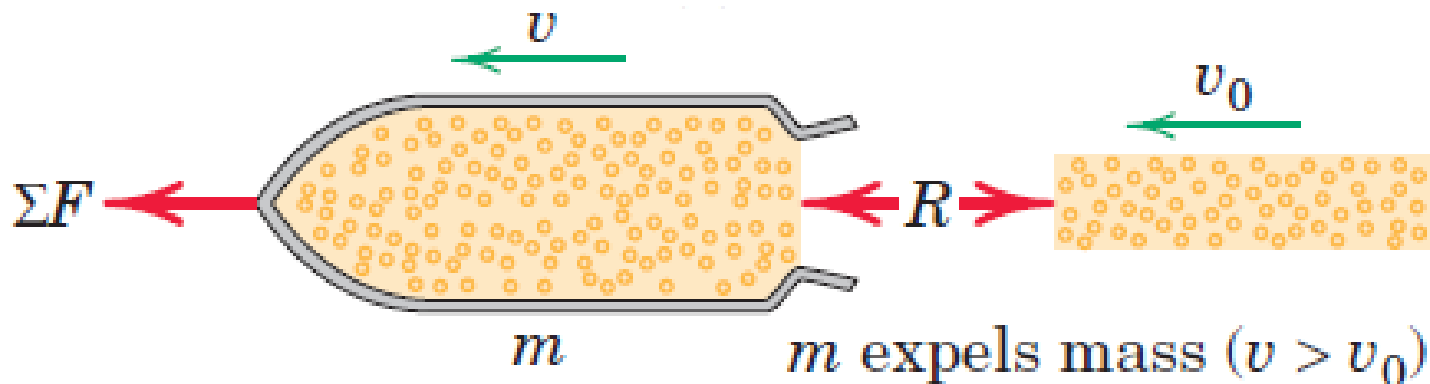


Variable Mass Flow

➤ When v_0 is zero,

$$\sum F = \frac{d(mv)}{dt} = \dot{m}v + m\dot{v}$$

$$\sum F = m\dot{v} + \dot{m}(v - v_0) = m\dot{v} + \dot{m}v$$

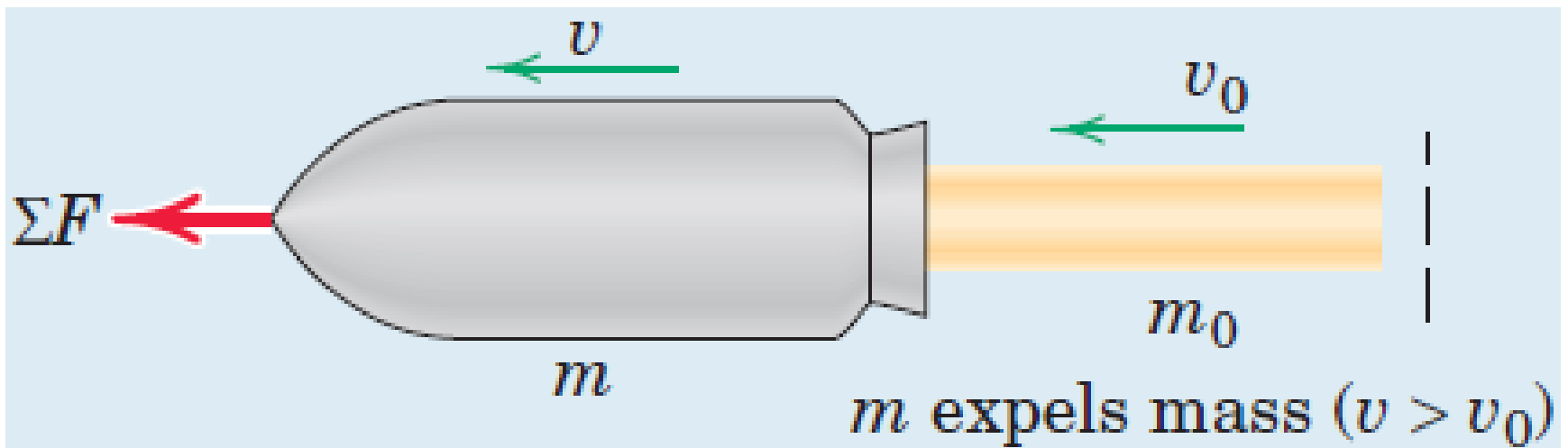


Variable Mass Flow

$$\sum F = \dot{G} = \frac{d(mv + m_0 v_0)}{dt} = \dot{m}v + m\dot{v} + \dot{m}_0 v_0 + m_0 \dot{v}_0$$

- Mass m_0 leaving mass m with velocity v_0
- Mass m_0 remains undisturbed after ejecting or leaving mass m
- Force exerted on the system is rate of change of linear momentum

$$\dot{v}_0 = 0$$
$$\dot{m}_0 = -\dot{m}$$



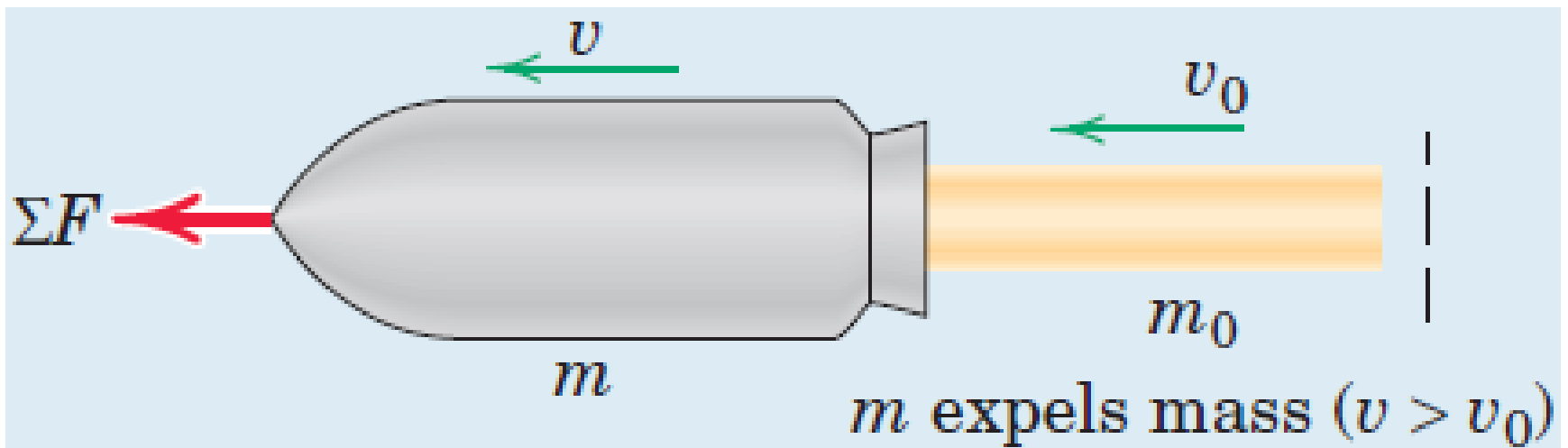
Variable Mass Flow

$$\sum F = \dot{G} = \frac{d(mv + m_0 v_0)}{dt} = \dot{m}v + m\dot{v} + \dot{m}_0 v_0 + m_0 \dot{v}_0$$

$$\dot{v}_0 = 0$$

$$\dot{m}_0 = -\dot{m}$$

$$\sum F = \dot{G} = \frac{d(mv + m_0 v_0)}{dt} = \dot{m}v + m\dot{v} - \dot{m}v_0 = \dot{m}(v - v_0) + m\dot{v}$$



Variable Mass Flow

$$\sum F = \dot{G} = \frac{d(mv + m_0 v_0)}{dt} = \dot{m}v + m\dot{v} + \dot{m}_0 v_0 + m_0 \dot{v}_0$$

$$\begin{aligned}\dot{v}_0 &= 0 \\ \dot{m}_0 &= -\dot{m}\end{aligned}$$

$$\sum F = \dot{G} = \frac{d(mv + m_0 v_0)}{dt} = \dot{m}v + m\dot{v} - \dot{m}v_0 = \dot{m}(v - v_0) + m\dot{v}$$

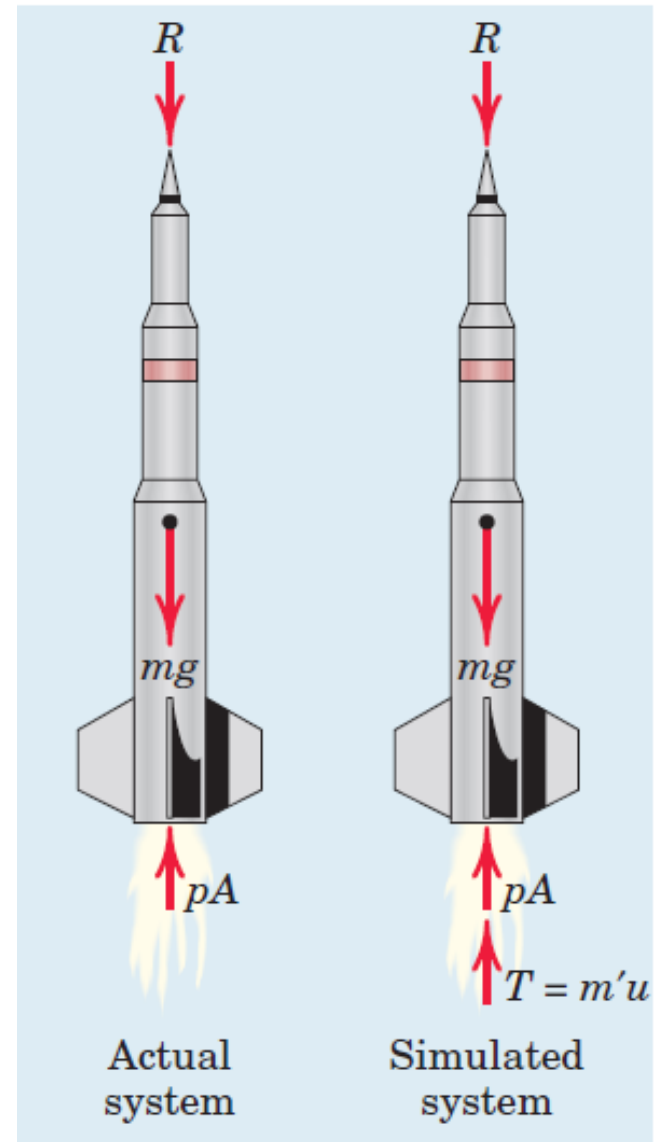
$$\sum F = \dot{G} = \dot{m}(v - v_0) + m\dot{v} = \dot{m}u + m\dot{v}$$

Rocket Propulsion

- Vertically ascending rocket
- System: mass within the volume, exterior surface of the rocket and the exit plane across the nozzle
- Forces acting on the system, instantaneous values of gravitational attraction mg , aerodynamic resistance R , and the force pA due to the average static pressure p across the nozzle exit plane of area A

$$\sum F = \dot{m}u + m\dot{v}$$

$$-mg - R + pA = \dot{m}u + m\dot{v}$$



Rocket Propulsion

$$\sum F = \dot{m}u + m\dot{v}$$

$$-mg - R + pA = \dot{m}u + m\dot{v}$$

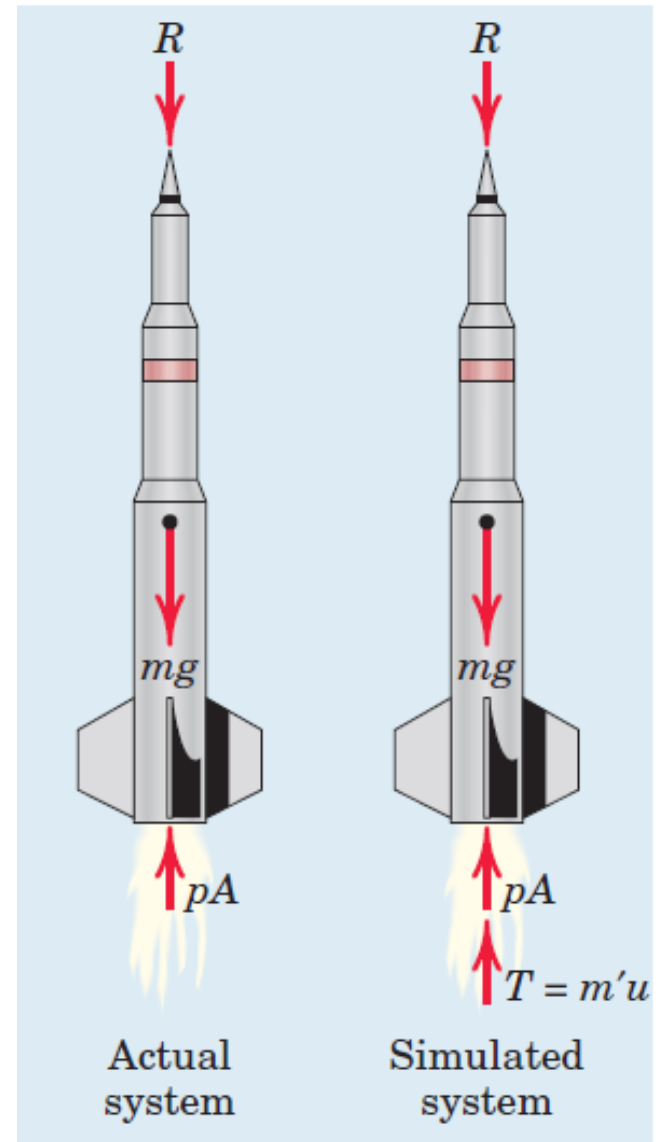
$$\dot{m} = -\dot{m}'$$

$$-mg - R + pA = -\dot{m}'u + m\dot{v}$$

$$-mg - R + pA + \dot{m}'u = m\dot{v}$$

$$-mg - R + pA + T = m\dot{v}$$

$$\sum F = ma$$



THANK YOU